

សាកលវិទ្យាល័យភូមិន្ទភ្នំពេញ

ROYAL UNIVERSITY OF PHNOM PENH

ប្រឆាំនបទរបាយការណ៍ស្រាវជ្រាវ

Explainable AI for Automated Grading of Khmer Language Short-Answer Questions in Education

A Final Report

In Partial Fulfilment of the Requirement for the Degree of
Bachelor of Engineering in Data Science and Engineering

Phork Norak

June 2026

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June 2026

សេចក្តីសង្ខេប

ការដាក់ពិន្ទុចម្លើយខ្លីដោយដៃត្រូវការពេលវេលាច្រើន ហើយចម្លើយដែលមានខ្លឹមសារដូចគ្នាអាចទទួលបានពិន្ទុខុសគ្នា។ ការសិក្សានេះស្វែងយល់ថា តើបញ្ញាសិប្បនិម្មិតអាចជួយគ្រូដាក់ពិន្ទុចម្លើយខ្លីជាភាសាខ្មែរ និងយល់ពីពិន្ទុដែលប្រព័ន្ធផ្តល់ឱ្យបានយ៉ាងដូចម្តេច។ ទិន្នន័យមានចម្លើយប្រឡងពិតប្រាកដចំនួន ១,១៨៤ ពីសិស្ស ២០៣ នាក់ នៅវិទ្យាល័យពីរក្នុងប្រទេសកម្ពុជា។ ចម្លើយទាំងនេះគ្របដណ្តប់លើសំណួរ ៤១ ក្នុងមុខវិជ្ជាជីវវិទ្យា ប្រវត្តិវិទ្យា ភូមិវិទ្យា និងផែនដីវិទ្យា។ ចម្លើយសរសេរដោយដៃ និងពិន្ទុរបស់គ្រូត្រូវបានបញ្ចូលជាទិន្នន័យឌីជីថល។ ដោយសារមិនមានចម្លើយគំរូផ្លូវការពីគ្រូ ចម្លើយរបស់សិស្សដែលទទួលបានពិន្ទុខ្ពស់បំផុតក្នុងសំណួរនីមួយៗ ត្រូវបានប្រើជាចម្លើយយោង។

ការសិក្សាប្រៀបធៀបម៉ូដែលបួនប្រភេទ គឺ TF-IDF ជាមួយ Support-Vector Regression, BiLSTM ជាមួយ Attention, Transformer encoder ពហុភាសា និង ក្រុមម៉ូដែលភាសា KhmerGrader។ ម៉ូដែល KhmerGrader ត្រូវបានបង្វឹកបន្ថែមលើទិន្នន័យដាក់ពិន្ទុភាសាខ្មែរដោយប្រើ QLoRA ដោយចាប់ផ្តើមពីម៉ូដែល Qwen, Gemma និង SEA-LION។ វិធីសាស្ត្រ SHAP បន្លិចពាក្យដែលមានឥទ្ធិពលលើពិន្ទុដែលម៉ូដែលផ្តល់ឱ្យ។ ពាក្យទាំងនេះត្រូវបានប្រៀបធៀបជាមួយចម្លើយយោង ដើម្បីពិនិត្យថាវាទាក់ទងនឹងខ្លឹមសារយោងកម្រិតណា។

ម៉ូដែល Qwen-KhmerGrader ផ្តល់ពិន្ទុស្របនឹងគ្រូបានល្អជាងគេ។ វាទទួលបានពិន្ទុវាស់ Quadratic Weighted Kappa (QWK) ចំនួន ០.៨៥០ បើប្រៀបធៀបនឹង ០.៨០២ សម្រាប់ម៉ូដែលបែបបុរាណ ០.៧៨៩ សម្រាប់ BiLSTM និង ០.៧៧៧ សម្រាប់ Encoder។ វាផ្តល់ពិន្ទុត្រូវគ្នាទាំងស្រុងនឹងគ្រូលើចម្លើយសាកល្បងចំនួន ៦៧% និងខុសពីគ្រូមិនលើសពីមួយពិន្ទុលើចម្លើយចំនួន ៨០%។ វាក៏ទទួលបានលទ្ធផលល្អជាងម៉ូដែលភាសាសម្រាប់ការប្រើប្រាស់ទូទៅទាំងបួនដែលទទួលបានតែសេចក្តីណែនាំដាក់ពិន្ទុដោយគ្មានឧទាហរណ៍ (zero-shot) (GPT-5.5, Claude Opus 4.8, Gemini 3.5 Flash និង DeepSeek V4)។ ក្នុងចំណោមម៉ូដែលទាំងនោះ Claude Opus 4.8 ទទួលបានលទ្ធផលល្អបំផុត ដោយមាន QWK ចំនួន ០.៧៣៤។

កម្មវិធីគំរូមួយត្រូវបានបង្កើតឡើង ដើម្បីបង្ហាញពិន្ទុ ពាក្យដែល SHAP បន្លិច និងមតិកែលម្អ។ លទ្ធផលបង្ហាញថា ម៉ូដែលទំហំតូចដែលបានបង្វឹកលើទិន្នន័យដាក់ពិន្ទុភាសាខ្មែរពិតប្រាកដអាចជួយគ្រូបាន។ ទោះជាយ៉ាងណា ការសិក្សានេះប្រើទិន្នន័យមានកម្រិត និងពិន្ទុពីគ្រូតែម្នាក់។ ចម្លើយយោងដែលជ្រើសពីចម្លើយរបស់សិស្សក៏អាចមិនពេញលេញដែរ។ ការពន្យល់របស់ប្រព័ន្ធ និងប្រយោជន៍ក្នុងថ្នាក់រៀននៅតែត្រូវការការវាយតម្លៃជាមួយគ្រូ។ គ្រូនៅតែជាអ្នកសម្រេចពិន្ទុចុងក្រោយ។

Keywords: Khmer short-answer grading; explainable AI; SHAP; KhmerGrader; teacher support.

ABSTRACT

Grading short answers by hand takes time, and similar answers can receive different marks. This study explores how AI can help teachers grade Khmer short answers and understand the suggested scores. It uses 1,184 real examination answers from 203 students at two Cambodian high schools. The answers cover 41 questions in Biology, History, Geography and Earth Science. The handwritten answers and teacher marks were entered into a digital dataset. Because official answer keys were unavailable, the highest-scoring student answer for each question was used as the reference.

The study compares four model families: TF-IDF with support-vector regression, BiLSTM with attention, multilingual Transformer encoders, and the KhmerGrader language models. KhmerGrader was trained further on Khmer grading data using QLoRA, starting from Qwen, Gemma and SEA-LION models. SHAP highlights words that affect each model's predicted score. The highlighted words are compared with the reference answer to check whether they point to relevant content.

Qwen-KhmerGrader gave the closest agreement with the teacher. Its Quadratic Weighted Kappa (QWK) score was 0.850, compared with 0.802 for the classical model, 0.789 for BiLSTM and 0.777 for the encoder. It matched the teacher's exact mark on 67% of test answers and was within one point on 80%. It also scored higher than four frontier language models: GPT-5.5, Claude Opus 4.8, Gemini 3.5 Flash and DeepSeek V4. These models were given grading instructions without worked examples (zero-shot). The best of those models, Claude Opus 4.8, reached a QWK of 0.734.

A prototype app shows the predicted score, SHAP word highlights and written feedback. The results suggest that a smaller model trained on real Khmer grading data can support teachers. However, the study uses a small dataset and marks from one grader. The reference answers chosen from students' work may also be incomplete. The explanations and classroom usefulness still need evaluation with teachers. The teacher remains responsible for the final mark.

Keywords: Khmer short-answer grading; explainable AI; SHAP; KhmerGrader; teacher support.

SUPERVISOR'S RESEARCH SUPERVISION STATEMENT

TO WHOM IT MAY CONCERN

Name of program: Honor Program of Data Science and Engineering

Name of candidate: Phork Norak

Title of research report: Explainable AI for Automated Grading of Khmer Language Short-Answer Questions in Education

This is to certify that the research carried out for the above-titled Bachelor's research report was completed by the above-named candidate under my direct supervision. This report material has not been used for any other degree. I played the following part in the preparation of this research report: guidance on the research direction, methodology, experimental design, and review of the written report.

Supervisor's name: Dr. Khim Chamroeun

Supervisor's signature: .....

Date: .....

CANDIDATE'S STATEMENT

TO WHOM IT MAY CONCERN

This is to certify that the research report that I, Phork Norak, hereby present, entitled *Explainable AI for Automated Grading of Khmer Language Short-Answer Questions in Education*, for the degree of Bachelor of Engineering in Data Science and Engineering at the Royal University of Phnom Penh, is entirely my own work and, furthermore, that it has not been used to fulfil the requirements of any other qualification in whole or in part, at this or any other University or equivalent institution.

No reference to, or quotation from, this document may be made without the written approval of the author.

Signed by (the candidate): .....

Date: .....

Signed by (the supervisor): .....

Date: .....

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CHAPTER 1
INTRODUCTION

1.1 Background to the Study

Assessment is a key part of education. It shows what students have learned, guides teaching, and helps students improve. Many assessments use *short-answer questions*. A student writes one to a few sentences, and the teacher checks the *correctness of the content* against a reference answer. These questions sit between multiple-choice questions, which are easy to mark but mostly test recall, and essays, which give more detail but take longer to grade. Short answers are common in school exams. Marking them by hand takes time and can be inconsistent, so teachers may have less time to give detailed feedback.

Automatic Short-Answer Grading (ASAG) studies how to score these answers with a computer. Early systems compared how similar the answer was to the reference. Later systems used deep neural networks, followed by Transformer-based and large language models [1, 2]. These methods have improved ASAG for languages with more available data, such as English, Arabic, and Chinese.

Those benefits have mostly *not* reached low-resource languages. **Khmer** (ភាសាខ្មែរ), the official language of Cambodia and spoken by more than 17 million people, is still low-resource in NLP. The Khmer script is also hard to process:

- it has **no spaces between words**, so even splitting a sentence into words is a machine-learning task in itself;
- it stacks subscript consonants (ើជើង) and reorders vowels inside a syllable;
- it contains invisible formatting characters that basic cleaning may miss.

Multilingual models see little Khmer during pre-training, and existing Khmer NLP work covers tokenisation, part-of-speech tagging, and search, but not grading [3, 4]. At the same time, AI in education is a growing priority for Cambodia [5], which makes a reliable, transparent Khmer grading tool both timely and useful.

Teachers also need to explain and justify a grade. A score without a reason is hard to trust or explain to a student. A classroom grading tool therefore needs to show *why* it gives a score. This study assesses each model in three ways: how closely its grades agree with the teacher's (accuracy), how often it gives the teacher's exact mark (deployment quality), and how well its explanation reflects its decision (explainability). A model may do well in one area and less well in another. We therefore report all three together rather than combining them into one ranking.

1.2 Problem Statement

We want to grade Khmer short answers automatically and clearly describe what the models can do with real classroom data. Three features of the data make this difficult:

- **No benchmark and no baselines.** There is no published Khmer ASAG dataset or system, so we must build both the benchmark and the reference results.
- **Different scoring scales.** Each question has its own maximum score, from the set $\{5, 6, 7, 8, 10, 12, 15, 20\}$. A raw score of "8" means different things on different questions, so the scores must be normalised.
- **A small, single-grader corpus.** The corpus has 1,184 answers graded by *one* teacher, so there is little data, no second grader to show how closely humans agree, and some inconsistent marks.

ASAG papers also often report two different kinds of performance together. *Research quality* means how closely the model's grades agree with the teacher's, measured by QWK. *Deployment quality* means whether the model predicts the *exact* mark in points. We report them separately because a model can do well on one and poorly on the other. Its grades may follow the same pattern as the teacher's, yet rarely match the exact mark. A teacher gains little time if every score needs to be checked again.

To make the task concrete, consider one real (translated) History item from the corpus. The **question** asks which factors caused Siam to break the citadel of Longvek; the **reference answer** lists several causes (the king's poor judgement, the loss of unity, and so on) and is worth a **maximum of 10 points**. One student wrote only "the factors that made Siam break Longvek: the loss of solidarity", naming a single correct cause. The teacher awarded **2 out of 10**. The grading system must read both texts, recognise that the answer is partially correct, and reproduce that partial mark. We put the raw score on a common scale by dividing by the maximum, $y = 2/10 = 0.2$, and into a five-class grade by rounding, $\ell = \text{round}(4 \times 0.2) = 1$. Every model in this study is trained to predict y for inputs of exactly this form, and is evaluated on both the resulting grade and the predicted raw mark.

1.3 Aim and Objectives of the Study

In this study, we aim to build the first reproducible, *explainable* ASAG benchmark for Khmer and to clearly describe what current model families can and cannot do with real classroom data. The objectives are:

- Collect and manually digitise real handwritten Khmer short-answer examination responses, together with the recorded teacher marks and question maxima.

- Normalise scores so questions with different maximum scores can share one grading and evaluation framework.
- Compare the Classical, RNN, Transformer and KhmerGrader families through a shared training and evaluation pipeline.
- Explain model predictions with SHAP word attribution and assess reference-overlap plausibility across all four families.
- Develop a prototype that returns a predicted score, readable word highlights and written feedback.

These objectives lead to the three research questions used in the final presentation:

- **RQ1 (Accuracy):** Which model family gives the best agreement with the human teacher?
- **RQ2 (Deployment):** Which model is most useful for classroom deployment based on exact mark and within-one-point agreement?
- **RQ3 (Explainability):** Can SHAP provide understandable word-level explanations across all model families?

We address RQ1 primarily with QWK, supported by accuracy and macro-F1; RQ2 with exact and within-one-point raw-mark agreement; and RQ3 with SHAP word highlights and reference-overlap plausibility. These measures show agreement with the recorded teacher marks and reference content. We still need to test directly whether teachers find the tool useful and its explanations understandable.

1.4 Rationale of the Study

This study addresses a clear gap: short-answer grading has improved for languages with more data, but not for Khmer. A grade can affect a student, so teachers need to be able to *check* how it was given. An explanation should point to the content behind the grade, allowing a teacher to compare it with the reference answer [6]. Explainability is therefore a main goal of this study. We use SHAP word attribution and check whether the highlighted words match the reference content.

The study makes the following contributions:

1. **The first Khmer ASAG benchmark**: 1,184 real teacher-graded answers (41 questions, 203 students, four subjects, two schools), documented with two curated dataset variants and an explicit label-noise case.

2. **A four-family comparison under one pipeline**: classical, RNN, Transformer, and a fine-tuned LLM (the KhmerGrader family).
3. **A cross-family SHAP explanation**: one model-agnostic SHAP word-attribution method applied to all families, with a plausibility (reference-overlap) check.
4. **A live prototype** that returns a score, a SHAP word-attribution explanation, and written feedback.

We collected examination answers from two Cambodian high schools, one in a province and one in the capital city, to include variation in student writing, answer quality and classroom context. These settings offer more variety than one classroom, but the sample is still small and local. The evaluation measures agreement with one grader's recorded marks. The two locations do not prove that the schools use different marking standards or that the results apply to all Cambodian schools.

1.5 Limitation and Scope

This study covers short-answer questions in Khmer from four secondary-school subjects (Biology, History, Geography, Earth Science). We study *content* scoring against a reference answer, not essay style or grammar. A single teacher graded every answer, so results are agreement *with that grader*, not against a measured level of agreement between human graders. The heavier models (the Transformer encoders and the LLM) need a GPU and were run on the MISTI-HPC cluster (an NVIDIA A40); the classical and RNN baselines run on the CPU. We revisit these limits in Chapter 6.

1.6 Structure of the Study

The rest of this report is organised as follows:

- **Chapter 2: Literature Review** reviews ASAG, the core concepts (Transformers and explainability), related work, and the research gap.
- **Chapter 3: Methodology** describes the research design, data collection, preprocessing, the four models, the tools, and the evaluation metrics.
- **Chapter 4: Experiment** describes the environment and the before-training, training, and testing phases, including the prototype.
- **Chapter 5: Results and Discussion** presents the results and discusses them.
- **Chapter 6: Limitations and Future Work** states the limits and the next steps.
- **Chapter 7: Conclusion** concludes the study.

CHAPTER 2
LITERATURE REVIEW

This chapter reviews the work this study builds on: an overview of automatic short-answer grading (Section 2.1), the challenges of Khmer (Section 2.2), the core concepts of Transformers and explainability (Section 2.3), related work (Section 2.4), and the gap in existing research (Section 2.5).

2.1 Overview of Automatic Short-Answer Grading

Automatic Short-Answer Grading (ASAG) scores a free-text student answer by comparing it, directly or indirectly, to a reference answer. The field has developed through four stages that overlap:

- **Similarity-based.** Early systems scored an answer by how closely it matched the reference, using surface features such as word overlap, n-gram and edit distance, and later dependency-graph alignment between the parse trees of the answer and the reference [7]. These methods are easy to inspect and need no training data. However, they focus on matching words and may miss answers that express the same idea in different words.
- **Shared tasks.** Public datasets and shared evaluation rules made it possible to compare systems. SemEval-2013 Task 7 [8] was especially influential. It introduced the key idea of testing on *unseen questions* and *unseen answers*.
- **Deep learning.** Researchers moved from hand-built features to features learned from data. These included word embeddings, recurrent networks, and Transformer encoders fine-tuned for scoring. They often expanded the training data to help with small datasets [1, 9]. These models can recognise meaning beyond matching words, but they need more data and are harder to explain.
- **Large language models.** The current stage fine-tunes or prompts billion-parameter generative models to produce a score directly from the question, reference, and answer. A common finding is that a model fine-tuned for grading often performs better than a larger general model used zero-shot [10], which is the design this study follows for its LLM pillar.

Throughout these stages, *Quadratic Weighted Kappa* (QWK) has stayed the standard metric for ordinal scoring, because it rewards near-misses and penalises large disagreements

more than small ones; it is usually reported alongside exact and adjacent (within-one-point) agreement so that a reader can separate overall correlation from exact-mark accuracy.

2.2 Challenges in Khmer

Khmer is a low-resource language whose script makes grading harder than in English:

- there are **no spaces between words**, so a system must first segment the text before any grading model sees it;
- the script stacks subscript consonants and reorders vowels inside a syllable, and it carries invisible characters that must be cleaned;
- multilingual models see little Khmer during pre-training.

Khmer NLP has focused on foundational tasks: word segmentation, part-of-speech tagging, and pre-trained models [3, 4]. This study uses the `khmer-nltk` segmenter for the segmentation step. More broadly, preparing Cambodian education for AI is an active concern [5], but grading Khmer student answers is still unaddressed.

2.3 Core Concepts: Transformers and Explainability

2.3.1 Transformers and Pre-trained Encoders

The Transformer [11] replaced recurrence with *self-attention* and is the backbone of modern NLP. Self-attention lets each token, or unit of text, use information from every other token in a sentence. For example, it can link "it" to the noun it refers to, or link an answer phrase to a similar phrase in the reference. A recurrent network passes this information step by step through hidden states. Self-attention connects all pairs of positions in one layer, helping the model connect words that are far apart as well as close together. Many of these calculations can run at the same time on a GPU.

Pre-trained encoders use this design to learn general text features from large amounts of unlabelled text. They can then be fine-tuned on a smaller task with labelled examples. BERT and its multilingual version mBERT [2] are trained with a masked-language-modelling objective (predict hidden words from their context); cross-lingual encoders such as XLM-R extend this to a hundred languages in one shared vocabulary, which is what makes them usable for a low-resource language like Khmer that has too little text to pre-train a dedicated model. However, these models see little Khmer during pre-training, so they learn weaker features for Khmer than for English. BERT, mBERT, XLM-R, GTE-multilingual, and the Qwen LLM used in this study are *all*

Transformers; they differ in size, training objective, and how they are adapted, not in their core self-attention design.

2.3.2 Explainability

Explainable AI (XAI) makes model decisions transparent. For text scoring the most useful family is *attribution*: assign each input word an importance score so a teacher can see which words drove the grade. Attribution methods fall into three groups:

- **Perturbation** methods change the input and watch the output: occlusion (leave-one-out) removes one word at a time, while LIME fits a simple local model to many random perturbations.
- **Gradient** methods (plain saliency, Integrated Gradients) use gradients to measure how changes in each input affect the output. They need a differentiable model and access to its internal calculations.
- **SHAP** (SHapley Additive exPlanations) distributes credit among the input words using Shapley values, which measure the average change in the prediction when a word is added to different groups of other words [12]. It is the field standard for explainable scoring and, in its model-agnostic form, treats the model as a black box, so one method works for a neural network, a Transformer, and the non-differentiable support-vector regressor alike.

This study uses SHAP as its explanation, following the explainable-scoring work of Kumar and Boulanger [13], who use SHAP to expose which features drive an essay score and show that such explanations can be useful in education. We judge an explanation by its *plausibility* [6]: whether the words it highlights look right to a teacher, measured as the overlap between the top SHAP words and the content of the reference answer. A plausible explanation highlights content that a teacher would check against the marking criteria. These highlights can then help the teacher give feedback to the student.

2.4 Related Work

In grading specifically, XAI is used both to explain scores and to give feedback. Kumar and Boulanger showed that explaining automated essay scores can be useful in education, using SHAP over interpretable features [13]. Closer to short-answer grading, the ExASAG framework pairs a transformer grader with post-hoc explanations so that each predicted score comes with a justification [14]. Most later scoring-XAI work has also relied on SHAP, LIME, or Integrated Gradients [15], and one study reports that SHAP and LIME

are poorly aligned with rubric assessment [16]. A systematic comparison of attribution methods for ASAG helps explain our choice of SHAP [17].

For evaluation, recent work compares fine-tuned models against prompted LLMs for grading and reports QWK with exact and adjacent agreement [15]. The closest comparable dataset is an Arabic ASAG corpus on a three-point scale, whose best Transformer reaches 95.67 %/77.22 % train/test accuracy and an unweighted Cohen's κ of 0.60 [18]; we use it as the anchor for honestly reporting the train/test gap.

Table 1 compares the approach of this study with selected earlier work and shows how the field has developed.

The early baselines establish the task and its metrics: Mohler et al. [7] score answers by semantic similarity and dependency-graph alignment, and the SemEval-2013 shared task [8] standardises evaluation and the unseen-question setting. The survey of Sung et al. [1] documents the shift from word embeddings to Transformers.

On the explainability side, Kumar and Boulanger [13] show that explaining essay scores can help education, but use SHAP over hand-built features, and Xu et al. [16] report that SHAP and LIME align only weakly with how a rubric is actually applied. Pinto and Shin [17] compare attribution methods for ASAG directly, including SHAP and leave-one-out, informing our use of SHAP.

On the model side, Chang and Ginter [10] test GPT-4 prompting and find fine-tuned specialists tend to lead, and Lachana [15] confirms this for Dutch ordinal scoring while reporting QWK with exact and adjacent agreement, the same set of measures used here. Latif and Zhai [19] show that a general LLM performs much better at scoring after fine-tuning than when only given a prompt; and Solano et al. [20] report that supervised fine-tuning of *open-source* LLMs narrows the gap to proprietary models for educational tools, scoring with greedy decoding (temperature 0) and reasoning disabled, the same inference setting we use. We extend this contrast to a low-resource language by comparing the fine-tuned KhmerGrader against current frontier models prompted zero-shot.

For Khmer specifically, Buoy et al. [3] provide pre-trained models and evaluation data but no grading task. Table 1 shows how the selected studies relate to our methods rather than comparing their scores.

Table 1: Representative studies in automatic short-answer grading.

Study	Language / data	Model approach	XAI method	Relevance to this study
Mohler et al. (2011) [7]	English short answers	Semantic similarity and dependency-graph alignment	None reported	Establishes an early similarity-based ASAG baseline
Dzikovska et al. (2013) [8]	English, SemEval-2013	Textual-entailment classification	None reported	Introduces standardised ASAG evaluation and unseen-question testing
Kumar and Boulanger (2020) [13]	English essays	Deep-learning essay scoring	SHAP	Demonstrates the educational value of explaining automated scores
Törnqvist et al. (2023) [14]	English short answers	Transformer-based grading	Post-hoc XAI	Provides a directly related explainable ASAG framework
Soulimani et al. (2024) [18]	Arabic short answers	Deep-learning and Transformer models	None reported	Provides a non-English ASAG reference for a comparatively low-resource setting
Lachana (2025) [15]	Dutch open-ended answers	Fine-tuned and prompted language models	None reported	Supports comparing fine-tuned and prompted LLMs using QWK and point-level agreement
Buoy et al. (2022) [3]	Khmer NLP datasets	Pre-trained Khmer language models	None reported	Establishes Khmer NLP resources but does not address grading
This study	**Khmer short answers**	**SVR, BiLSTM, Transformer, and fine-tuned LLM**	**SHAP**	**Introduces a Khmer ASAG benchmark and applies one explanation method across four model types**

Note: The studies use different datasets, languages, tasks, and scoring scales. The table summarises methodological relevance rather than numerical performance.

Table 1 shows grading studies in several languages and with several model families. However, most provide no word-level explanation. The studies that use SHAP apply it to one model type, and none targets Khmer. This is the gap addressed here.

2.5 Gap in Existing Research

Two gaps stand out:

1. there is **no published ASAG benchmark for Khmer**, so the field has no data and no baselines;

2. most ASAG work provides no word-level explanation, and the explainable studies apply SHAP to a single model type rather than across different model families.

This study addresses both: it builds the first reproducible Khmer ASAG benchmark and applies one model-agnostic SHAP explanation across four different model families (including the non-differentiable classical model).

CHAPTER 3
METHODOLOGY

3.1 Research Design

We use experiments and numerical measures to compare three things: agreement with a human grader, accuracy of raw marks for classroom use, and how plausible the explanations are. All four model families are evaluated on the same normalised and ordered grade scales, but they learn in different ways. SVR uses a regression objective. Neural regression heads minimise mean-squared error. The generative KhmerGrader models learn to produce an integer score. For evaluation, predictions are expressed as normalised scores in $[0, 1]$ and converted to grades or raw marks for each question. Figure 1 shows the full research process.

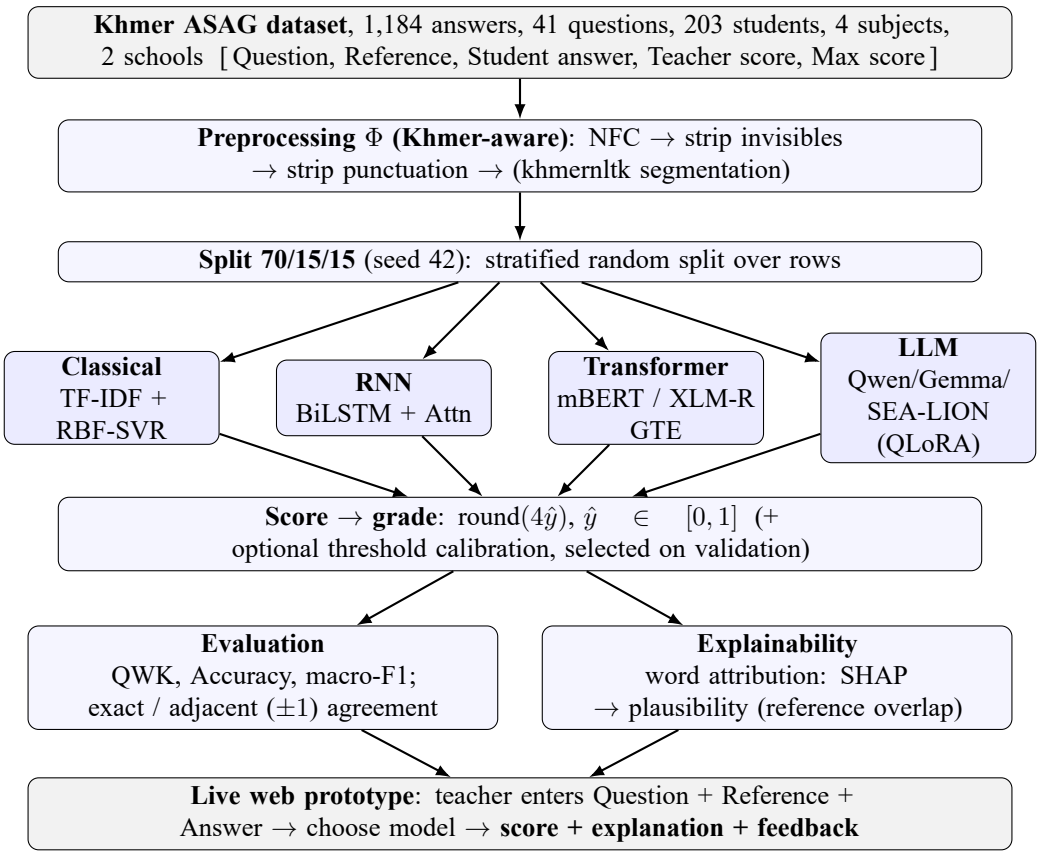


Figure 1: Overall research pipeline of this study.

3.2 Data Collection

3.2.1 Provenance

The dataset contains real Khmer short answers from semester 1 examinations in the 2026 school year. The source material was handwritten examination papers collected from two Cambodian high schools, one in a province and one in the capital city. Students wrote the responses and a teacher assigned their marks. The study did not generate synthetic answers.

Collection and preparation followed five stages, illustrated in Figure 2:

1. Collect the handwritten examination papers from the two schools.
2. Record the student answers and the teacher's awarded marks.
3. Manually type the handwritten Khmer answers into text that a computer can read.
4. Organise the question, reference answer, student answer, awarded score and question maximum into structured records.
5. Apply the shared cleaning and dataset preparation procedure before model training and evaluation.

The two schools add variety in writing styles and classroom settings, but the dataset does not represent all schools in the country. A single experienced high-school subject teacher supplied the recorded grading decisions, so the evaluation measures agreement with that grader.

Reference-answer construction. An official teacher-provided answer key was unavailable. For each question, the highest-scoring available student response was therefore selected as the reference answer. This gives an example of relevant content, but it is not a checked, complete marking guide. The highest student mark may be below the question maximum. The selected reference can therefore miss correct points or contain errors. To normalise a score, we always divide by the question maximum, not by the highest student mark.

3.2.2 The Corpus

The corpus has **1,184** short answers (after dropping one incomplete row) covering **41** distinct questions across four subjects (Biology, History, Geography, Earth Science), written by **203** students. Each record holds the question, a reference answer, the student's answer, the teacher's integer score, and the question's maximum score, which varies across

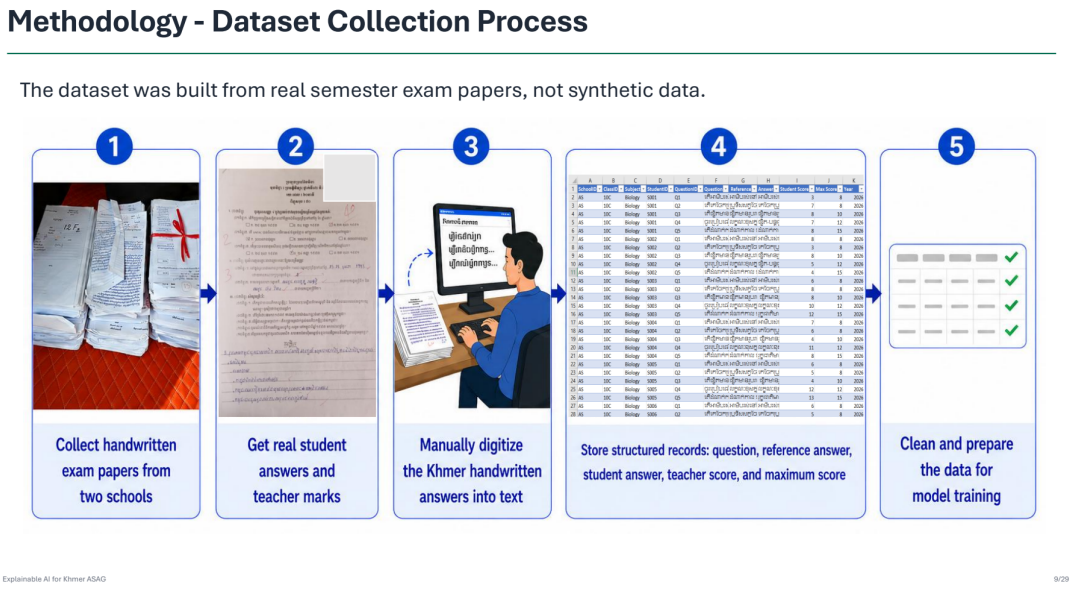


Figure 2: Collection and preparation of the real examination corpus: handwritten papers, teacher-marked responses, manual Khmer transcription, structured records and data preparation. The process illustration is reproduced from the final presentation.

$\{5, 6, 7, 8, 10, 12, 15, 20\}$. Table 2 shows one real record exactly as stored. Table 3 then summarises the full dataset, and Figures 3 and 4 show how its main features are distributed.

To place all questions on a common scale, we give each answer a *normalised score*, defined as:

$$y = \frac{\text{Student Score}}{\text{Max Score}} \in [0, 1]. \tag{3.1}$$

Each normalised score is then converted to one of five ordered *grade labels*, defined as:

$$\ell = \text{clip}\big(\text{round}(4y),\ 0,\ 4\big) \in \{0, 1, 2, 3, 4\}, \tag{3.2}$$

where the factor 4 maps the interval $[0, 1]$ onto the five integer grades. Table 3 shows that many labels are full-credit grades. Earth Science has only nine answers. We keep it to include the subject, but recognise that it has much less data than the others.

The four panels in Figures 3 and 4 show important features of the dataset. Each helps explain the later analysis. The subject panel (Figure 3, left) shows that Biology, History, and Geography are well represented while Earth Science has very few answers, which is why we treat Earth Science cautiously. The max-score panel (right) shows that 10- and 15-point questions dominate, so the score scale differs across questions and must be normalised. The grade-label panel (Figure 4, left) shows the heavy skew toward full credit (label 4), which is why we report macro-F1 alongside accuracy. The answer-length panel (right) shows most answers are short, with a median near 193 characters and a few much longer answers.

Table 2: A real corpus record, shown verbatim (Biology; school AS, class 11K, student S056, question Q7). The student answer is vague and only partially correct, so the teacher awarded 2 of 5 points. English glosses are added for the reader and are not part of the data.

Field	Content
Question	ហេតុអ្វីបានជាមនុស្សស្រូបយកខ្យល់ដែលមានកាបូនម៉ូណូអុកស៊ីតអាចបង្កគ្រោះថ្នាក់ដល់ស្លាប់បាន ? *(Why can breathing air that contains carbon monoxide be fatal to humans?)*
Reference	បានជាមនុស្សស្រូបយកខ្យល់ដែលមានកាបូនម៉ូណូអុកស៊ីដអាចបង្កឲ្យមានគ្រោះថ្នាក់ដល់ស្លាប់បានព្រោះកាបូនម៉ូណូអុកស៊ីដមានសមត្ថភាពចាប់យកក្នុងឈាមបានខ្លាំងជាងអុកស៊ីសែន វារារាំងឈាមមិនឲ្យដឹកនាំអុកស៊ីសែនទៅចិញ្ចឹមសរីរាង្គជាពិសេសខួរក្បាលដែលបណ្តាលឲ្យស្លាប់និងសន្លប់។ *(Because carbon monoxide binds in the blood more strongly than oxygen and blocks the blood from carrying oxygen to nourish the organs, especially the brain, causing unconsciousness and death.)*
Student answer	បានជាមនុស្សស្រូបយកខ្យល់ដែលមានកាបូនម៉ូណូអុកស៊ីដអាចបង្កគ្រោះថ្នាក់ដល់ស្លាប់បានព្រោះអុកស៊ីតដែលមនុស្សត្រូវការ ឧស្ម័នកាបូនិចនិងទឹក ទើបអាចរស់បាន។ *(Because the oxygen that humans need, carbonic gas and water, is what lets them live.)*
Score / Max	2 / 5 (normalised $y = 0.40$; grade label $\ell = \text{round}(4 \times 0.40) = 2$)

Table 3: Khmer ASAG corpus statistics at a glance.

Property	Value	Notes
Answers	1,184	one incomplete row dropped
Questions	41	small question pool
Students	203	two schools (province + city)
Subjects	4	Biology 435, History 433, Geography 307, Earth Sci. 9
Max scores	$\{5, 6, 7, 8, 10, 12, 15, 20\}$	most common: 10 (404), 15 (306), 8 (135), 5 (130)
Grade labels (0–4)	14 / 153 / 327 / 191 / 499	skewed: ≈42 % receive full credit (label 4)
Answer length (chars)	median 193 (IQR 140–383)	min 34, max 1,477, mean 278.7

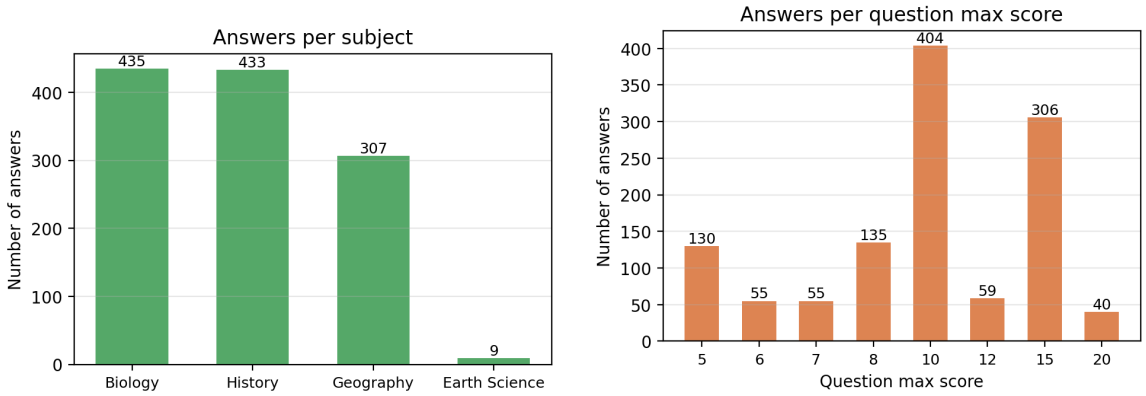


Figure 3: Number of answers per subject (left) and per question maximum score (right). Biology, History, and Geography dominate; Earth Science is small; and the 10- and 15-point questions are the most common, so the scoring scale differs across questions.

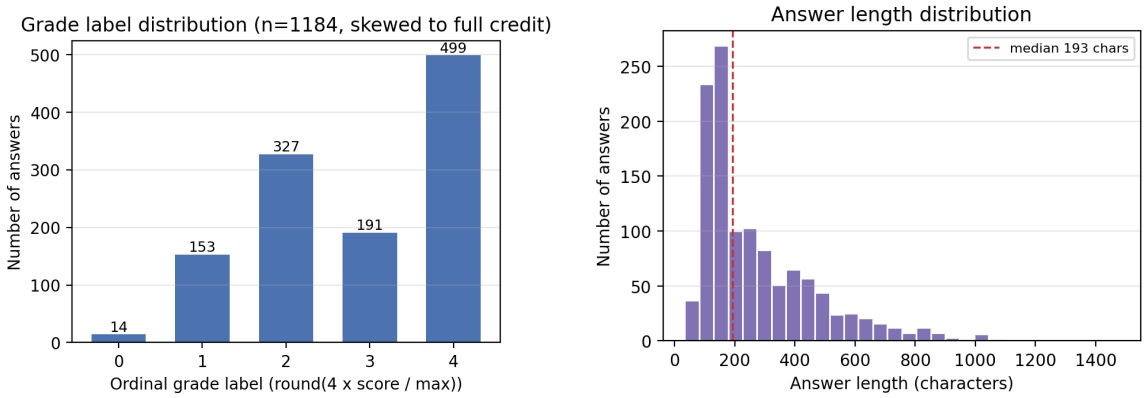


Figure 4: Grade-label distribution (left) and answer-length distribution (right). Labels are heavily skewed toward full credit (label 4), and most answers are short (median near 193 characters), with a few much longer answers.

All experiments train and evaluate on a single curated dataset, `no10c` (909 answers): the 1,184 collected answers minus one subset, class 10C Biology, whose recorded marks show clear inconsistencies (Section 3.2.3). The dataset keeps the full five-class label range (grades 0–4).

Unless stated otherwise, we split the data 70/15/15 into training, validation, and test sets with a fixed seed (42), using a *stratified random* split over rows.

3.2.3 A Real Case of Label Noise: Class 10C Biology

The `no10c` dataset removes a group of answers with clear grading inconsistencies. It is not a random reduction of the data. Looking at class 10C, Biology, Question 2 (maximum score 8), the 55 student records contain only **seven different answer texts**. However, *identical* answers received marks across almost the full score range (Table 5). One and the same answer text was scored 3, 4, 5, 6, 7, and 8. The same input has six different

labels, so a model cannot learn one consistent rule from it. Removing this group therefore improves every model. Removing it raises the classical champion's QWK by about $+0.044$ ($0.774 \rightarrow 0.818$). This gain is similar to or larger than the gaps between model designs. In contrast, *refining* the cleaning of the same data by removing zero-width and formatting characters changes classical QWK by at most 0.003. The main gain comes from *which data* is kept, rather than small changes to text cleaning (Section 5.1.4).

Table 4: Two curated dataset variants.

Variant	Definition	Rows
`full`	Original corpus	1,184
`no10c`	Drop the noisy "10C Biology" subset	909

Table 5: Grader inconsistency in class 10C Biology, Question 2 (55 students, 7 distinct answer texts, max score 8).

Distinct answer text	Different scores awarded to that identical text
A	3, 8
B	6, 7
C	3, 5, 6
D	3, 4, 7, 8
E	3, 4, 5, 6, 7, 8
F	5, 7, 8
G	3, 4, 5, 6, 7

Table 6 shows five of these records taken directly from the source spreadsheet. The clearest evidence is that *word-for-word identical* answers were given different scores: records 92 and 97 carry the same text but were scored 7 and 4, and records 102 and 107 carry the same text but were scored 4 and 7.

Real classroom variation. The records include identical answers assigned different marks, responses that repeat the question, and zero-score responses with no meaningful answer content. Biology, History and Geography contribute most records, while Earth Science has limited coverage. Grades are skewed toward high marks. We retain grade 0 and report macro-F1 alongside accuracy. Removing the class-10C Biology group does not prove that all remaining marks are consistent.

3.2.4 Data Governance

The corpus contains real student work by minors. We obtained access by approaching each school and getting permission directly from the school Director; no

Table 6: Real records from the source spreadsheet (Class 10C Biology, Question 2; max score 8). Records 92 and 97 are identical yet scored 7 and 4; records 102 and 107 are identical yet scored 4 and 7. The same answer received different labels, so no model can learn one consistent rule from this group.

Question 2 (max score 8): តើគេចែកប្រូទីសជាប៉ុន្មានក្រុម? អ្វីខ្លះ?

No.	Student answer	Score
92	គេចែកប្រូទីសសត្វជា ៤ប្រភេទគឺ-ប្រូទីសផ្លាស់ទីដោយជើងបញ្ឆោត : អាមីប-ប្រូទីស ផ្លាស់ទីដោយរោមញ៉ែរ : ប៉ារ៉ាមេស៊ី-ប្រូទីសផ្លាស់ទីដោយ ប្លាសែល-ប្រូទីសផ្លាស់ទីដោយ ស្ប៉ូ។	7
97	គេចែកប្រូទីសសត្វជា ៤ប្រភេទគឺ-ប្រូទីសផ្លាស់ទីដោយជើងបញ្ឆោត : អាមីប-ប្រូទីស ផ្លាស់ទីដោយរោមញ៉ែរ : ប៉ារ៉ាមេស៊ី-ប្រូទីសផ្លាស់ទីដោយ ប្លាសែល-ប្រូទីសផ្លាស់ទីដោយ ស្ប៉ូ។	4
102	គេចែកប្រូទីសសត្វជា ៤ ក្រុម គឺ -ណ្វៃយ៉ូ -ជើងបញ្ឆោតអាមីប -ស៊ីគូប្លាស -វ៉ាគុយអូលកន្ត្រាក់	4
107	គេចែកប្រូទីសសត្វជា ៤ ក្រុម គឺ -ណ្វៃយ៉ូ -ជើងបញ្ឆោតអាមីប -ស៊ីគូប្លាស -វ៉ាគុយអូលកន្ត្រាក់	7
177	គេចែកប្រូទីសសត្វជា ៣ ក្រុម -ប្រូទីសសត្វ -ប្រូទីសរុក្ខជាតិ - និងប្រូទីសផ្សិត។	7

formal IRB process applies at this level in this setting. We store only coded identifiers, with no student names. We use the data only to research a tool that helps a teacher, not one that grades independently. One teacher graded all answers, so the system learns that teacher's decisions. We report agreement *with this grader*. To reduce part of the problem, we document and remove the group with clear grading inconsistencies before training (Section 3.2.3).

3.3 Data Preprocessing

Khmer is written without spaces between words, and the same word can be stored using different byte sequences (for example stacked versus unstacked syllables), so the text needs careful cleaning before a model sees it. We apply the following steps, in order: NFC normalisation; removal of *invisible* characters (zero-width spaces, control and format characters, and bullet markers such as ● and ○); punctuation stripping; and finally word segmentation with `khmer-nltk`. We keep digits and letters as answer content.

Khmer can write the same word in stacked or unstacked form, giving one word several byte sequences; we rely on the segmenter to identify word boundaries and do not add a separate character-normalisation step. This keeps the processed text readable. Every

heatmap and piece of feedback shown to a teacher uses these readable text units directly.

3.3.1 Model Input Representation

The preprocessing pipeline produces two consecutive stages of the same text: the `clean` stage (NFC normalisation, invisible-character removal, and punctuation stripping) and the `segment` stage (`clean` plus `khmer-nltk` word segmentation). Each model uses the stage that suits its design: the classical TF-IDF model reads the *segmented* text, because explicit word boundaries help a bag-of-n-grams, while the BiLSTM, the Transformer encoder, and the LLM read the *cleaned* text directly, since they learn their own units from the text stream itself. The model input is the pair (answer, reference); the LLM additionally reads the question, which it can actually use. We selected these input forms using the validation set rather than choosing them by hand. Section 5.1.4 shows that the alternatives change the results only slightly. The segmented tokens are also the word units used by every SHAP explanation, so all highlights shown to a teacher are human-readable words.

3.3.2 An End-to-End Walkthrough of One Answer

To show how the full process works and can be repeated, we follow the student answer from the example record (Table 2) through every stage, from the raw text to the final grade and explanation. The `clean` stage applies fixed rules to remove the Khmer full stop ។ and any invisible characters and normalises spacing; the `segment` stage then runs `khmer-nltk`, which inserts the word boundaries that written Khmer omits, giving the human-readable tokens that every word-attribution heatmap displays (Chapter 5; Appendix F). The grader awarded $2/5$, so the supervised target is the normalised score $y = 0.40$ and the grade label $\ell = \text{round}(4 \times 0.40) = 2$. When making a prediction, a model outputs $\hat{y} \in [0, 1]$; the reported raw mark is $\text{round}(\hat{y} \times 5)$ and the class label is $\text{round}(4\hat{y})$. For the explanation, SHAP distributes the predicted score across the segmented tokens, producing the heatmap a teacher sees in the prototype.

Table 7: Preprocessing modes and input formats swept by every experiment.

Axis / value	Definition
`raw`	strip invisibles + strip whitespace
`clean`	`raw` + strip punctuation
`segment`	`clean` + `khmer-nltk` word segmentation
`ra`	input pair (answer, reference)
`qar`	input pair (question + answer, reference)

Table 8: One answer (from Table 2) traced through the pipeline. The original table body is reproduced from the latest final report to preserve Khmer shaping and the displayed preprocessing example.

Stage	Result
`raw`	បានជាមនុស្សស្រូបយកខ្យល់ដែលមានកាបូនម៉ូណូអុកស៊ីដអាចបង្កគ្រោះថ្នាក់ដល់ស្លាប់បានព្រោះអុកស៊ីតដែលមនុស្សត្រូវការ ឧស្ម័នកាបូនិចនិងទឹក ទើបអាចរស់បាន។
`clean`	បានជាមនុស្សស្រូបយកខ្យល់ដែលមានកាបូនម៉ូណូអុកស៊ីដអាចបង្កគ្រោះថ្នាក់ដល់ស្លាប់បានព្រោះអុកស៊ីតដែលមនុស្សត្រូវការ ឧស្ម័នកាបូនិចនិងទឹក ទើបអាចរស់បាន (។ and invisibles removed)
`segment`	បាន ជា មនុស្ស ស្រូប យក ខ្យល់ ដែល មាន … (word boundaries inserted by `khmer-nltk`)
target	$y = 2/5 = 0.40$; grade label $\ell = \text{round}(4 \times 0.40) = 2$
inference	model emits $\hat{y}$; raw mark $= \text{round}(\hat{y} \times 5)$, label $= \text{round}(4\hat{y})$
explanation	SHAP spreads the predicted score over the segmented tokens $\rightarrow$ heatmap

3.4 Models Used

All models use the same answer/reference text pair, optionally with the question added first. They output $\hat{y} \in [0, 1]$ and are trained to minimise mean-squared error against the normalised target y.

3.4.1 Classical: TF-IDF with RBF-SVR

We represent each text with character-level TF-IDF features (`char_wb` n-grams of length 2–4, capped at 15,000 features), which suit Khmer because they do not depend on perfect segmentation. For an answer/reference pair with feature vectors $\mathbf{a}$ and $\mathbf{b}$, we form the following interaction vector:

$$\big[\, \mathbf{a};\ \mathbf{b};\ |\mathbf{a} - \mathbf{b}|;\ \mathbf{a} \odot \mathbf{b};\ \cos(\mathbf{a}, \mathbf{b}) \,\big], \tag{3.3}$$

where ; joins the vectors, $|\mathbf{a} - \mathbf{b}|$ the element-wise absolute difference, $\odot$ the element-wise product, and cos the cosine similarity. We feed this vector to a support-vector regressor with an RBF kernel and penalty $C = 1$. This model trains in seconds on a CPU.

We also test two simpler classical baselines. The `tfidf_cos` scorer directly uses the cosine similarity between the answer and reference TF-IDF vectors, with no training. The `fasttext_cos` scorer does the same with 100-dimensional fastText embeddings. Both score below the support-vector regressor. This shows that learning the regression helps beyond using TF-IDF features alone. RBF-SVR is the classical champion reported throughout.

3.4.2 RNN: Character BiLSTM with Attention

We embed each text and encode it with a bidirectional LSTM. An additive attention layer combines the hidden states at each position $\mathbf{h}_t$ into one vector:

$$\alpha_t = \text{softmax}\big(\mathbf{w}^\top \tanh(W\mathbf{h}_t)\big), \qquad \mathbf{e} = \sum_t \alpha_t \mathbf{h}_t, \tag{3.4}$$

where W and $\mathbf{w}$ are learned, α_t is the weight for position t, and $\mathbf{e}$ is the pooled encoding. We encode the answer and reference separately and combine them through the same four-way interaction before a small MLP head with a sigmoid output. We limit the vocabulary at the 5,000 most frequent entries, use 128-dimensional embeddings, two bidirectional LSTM layers with 128 hidden units per direction, and dropout 0.3, and train with the Adam optimiser at learning rate $1\text{e}{-3}$, batch size 64, for up to 20 epochs with early stopping (patience 4) on validation QWK. The attention is part of the architecture; we do *not* use it as an explanation (we use SHAP, Section 3.6.2).

3.4.3 Transformer: Multilingual Encoders

We fine-tune multilingual encoders, mBERT, XLM-R, and GTE-multilingual [2], in two configurations: a *dual* encoder that embeds the answer and reference separately and then combines them with the four-way interaction above, and a *cross* encoder that feeds both texts into one sequence so that self-attention can compare them token by token. We keep the bottom six Transformer layers fixed. These layers contain general features shared across languages, and fixing them helps reduce over-fitting on a small dataset. We limit inputs to 256 tokens and fine-tune the other layers and a regression head. We use the AdamW optimiser at learning rate $2\text{e}{-5}$, weight decay 0.01, dropout 0.2, and batch size 16. Training runs for up to 20 epochs, with early stopping (patience 4) based on validation QWK. GTE-multilingual has an unstable cache for rotary position values that causes invalid (NaN) outputs. We fix this by rebuilding the cosine cache in full (fp32) precision when loading the model.

3.4.4 LLM: KhmerGrader Family with QLoRA

We fine-tune three open language models to form the KhmerGrader family: Qwen-KhmerGrader-4B, Gemma-KhmerGrader-4B and SEA-LION-KhmerGrader-E2B. Their base sizes are 4B, 4B and approximately 2.3B parameters, respectively. Qwen is the family champion used in the main four-pillar comparison; the family comparison includes all three models (Table 12). Each uses QLoRA [21]. This trains a small set of low-rank adapter weights while keeping the base model fixed. The base uses 4-bit quantisation (NF4, double-quantised) to reduce memory use, making it possible to fine-tune a billion-parameter

model on one GPU. The adapters use rank 16, $\alpha = 16$, no adapter dropout, and target the attention and MLP projections (q, k, v, o, gate, up, down). We give grading instructions in a fixed prompt. It lists the score scale, question, reference and student answer, and asks for one integer. Training uses completion-only loss: prompt tokens are ignored when calculating the loss, so the model learns from the score token only:

```
Below is a Khmer short-answer grading task. Score the student's
answer on a scale from 0 to {Max}.
Question: {question}
Reference answer: {reference}
Student answer: {answer}
The score (integer from 0 to {Max}): {score}
```

Unlike the other pillars, the LLM also reads the *Question* line, since a generative model can actually use the question context. For instruction-tuned bases, we use the model's chat template and turn reasoning off. The output is therefore just an integer rather than a written chain of thought. At inference we use **greedy decoding (temperature 0)** with reasoning disabled, the same configuration recent work adopts for fine-tuned open-source scoring models [20]: sampling temperature in the 0–1 range affects output variance rather than mean accuracy, and temperature 0 makes the output easier to repeat. This matters because we report one fixed score per answer. We then extract the first integer from the output and limit it to $[0, \text{Max}]$.

For training, we fine-tune each base on the `no10c` training split, for up to 10 epochs at batch size 4 with gradient accumulation to an effective batch of 16, a 5 % warm-up, and a validation-selected learning rate in $\{1, 2, 3\} \times 10^{-4}$, keeping the epoch with the best validation QWK.

The LLM pillar therefore has two settings: the *untuned base* prompted zero-shot (the in-pillar baseline) and its *QLoRA-fine-tuned* counterpart, the KhmerGrader family. Comparing the two shows the value of fine-tuning, following the fine-tuning-for-scoring approach of Latif and Zhai [19]; the comparison is reported in Chapter 5.

3.4.5 Frontier-Model Baselines

To test whether a small fine-tuned specialist can match much larger general models on Khmer, we add a *frontier baseline*: four proprietary or open frontier LLMs (a GPT, a Claude, a Gemini, and a DeepSeek model) prompted on the same test answers with the *same* grading prompt as the KhmerGrader. Each is tested *zero-shot*, meaning without worked examples, in two settings. One asks for an integer directly. The other allows reasoning before we extract the final integer. This gives the frontier models a chance to use their

stronger setting. The score is parsed and clipped exactly as for our own LLM. We grant the frontier models the reasoning variant precisely because we cannot fine-tune them; our own models are scored without reasoning, since the fine-tuned KhmerGrader emits a bare integer by design and, of the three open bases, only Qwen exposes a reasoning mode (the Gemma and the Gemma-based SEA-LION bases do not). This is also consistent with prior work, where chain-of-thought has not reliably improved ordinal scoring and adds latency. Few-shot prompting of all models is left to future work (Section 6.2).

We use closed APIs only for a one-off *evaluation* comparison. The prototype does not need them to run; its language-model feedback always uses an open model. Closed services are continuously updated, so we **pin each model version and record the evaluation date**, release the exact prompt and parser, and **cache every raw API response**; the cached snapshot makes these numbers reproducible even if a provider later revises the model. We state plainly that closed-model scores are a dated snapshot, whereas KhmerGrader weights, code, and data remain fully reproducible.

3.4.6 From Score to Grade

We map the continuous prediction to an integer grade by rounding $4\hat{y}$ (for the five-class label) or $\hat{y} \times$ Max Score (for the raw integer score). For example, a prediction $\hat{y} = 0.72$ on a question worth 10 points becomes raw score round$(0.72 \times 10) = 7$ and grade label round$(4 \times 0.72) = 3$. We also test two low-cost changes to this score conversion:

- **The max-score feature.** Because "0.5" means 5/10 on one question but 10/20 on another, we give the model the question's maximum score as an extra input feature, normalised by the largest observed maximum (20) so it lands in $(0, 1]$; this gives consistent deployment gains.
- **Post-hoc threshold calibration.** This adjusts the four boundaries that convert $\hat{y}$ to the five classes, choosing the boundaries with the highest QWK on the validation set. We treat calibration as an ablation, not part of the headline, because (as Chapter 5 shows) it helps the classical model but hurts the BiLSTM, so all headline numbers are uncalibrated.

3.5 Tools and Technologies

3.5.1 Programming Language

All work uses Python.

3.5.2 Frameworks and Libraries

- **scikit-learn**: TF-IDF features, the support-vector regressor, the train/validation/test splits, and every evaluation metric (QWK, accuracy, F1, exact/adjacent agreement).
- **PyTorch**: the BiLSTM with attention.
- **Hugging Face Transformers**: the multilingual encoders (mBERT, XLM-R, GTE).
- **unsloth** with **QLoRA**: efficient 4-bit fine-tuning of the KhmerGrader family.
- **khmer-nltk**: Khmer word segmentation in the preprocessing pipeline.
- **pandas** and **NumPy**: loading the corpus and managing features and predictions.
- **matplotlib**: the result figures.
- **Gradio**: the interactive web prototype.

3.5.3 Execution Environment

We run all experiments as Python scripts inside `tmux` sessions on a node of the MISTI-HPC cluster (the Ministry of Industry, Science, Technology and Innovation High-Performance Computing facility): a virtualised host with 32 CPU cores, 64 GiB of RAM, an NVIDIA A40 GPU (48 GB GDDR6 VRAM), and 536 GB of disk.

3.6 Model Evaluation Metrics

3.6.1 Agreement and Deployment Metrics

We evaluate every model on two levels: *research* metrics that measure how well the predicted grades agree with the teacher's (QWK, Cohen's κ, accuracy, macro-F1), and *deployment* metrics that measure how often the model gives a usable integer mark in the classroom (exact and adjacent agreement). Each metric is defined below.

Quadratic Weighted Kappa (QWK). Our primary metric is *Quadratic Weighted Kappa*, the field standard for ordinal scoring. It is calculated using three parts: the confusion matrix O between the predicted and true grade labels, the expected matrix E under chance agreement, and the quadratic weight matrix $w_{ij} = (i-j)^2/(K-1)^2$, where K is the number of grade classes. QWK is then defined as:

$$\text{QWK} \;=\; 1 - \frac{\sum_{i,j} w_{ij}\, O_{ij}}{\sum_{i,j} w_{ij}\, E_{ij}}. \tag{3.5}$$

A QWK of 1 is perfect agreement and 0 is chance agreement; because the weights grow with the square of the distance $(i - j)$, large disagreements are penalised much more than small ones.

Cohen's kappa. For the comparison with the Arabic anchor [18] we also report the *unweighted* Cohen's κ, which treats every disagreement equally regardless of its size.

Accuracy and macro-F1. These two metrics judge the five-class grade labels directly. They are defined as:

$$\text{Acc} = \frac{1}{N}\sum_{n=1}^{N} \mathbf{1}[\hat{\ell}_n = \ell_n], \qquad \text{macro-F1} = \frac{1}{K}\sum_{c=1}^{K} \frac{2P_c R_c}{P_c + R_c}, \tag{3.6}$$

where $\mathbf{1}[\cdot]$ equals one when the condition is true and zero otherwise. P_c and R_c are the precision and recall of class c. Macro-averaging gives each grade equal weight. This matters because some grades are much more common than others.

Exact and adjacent agreement. For deployment we report *exact agreement* (the predicted integer score equals the teacher's mark) and *adjacent agreement* (the predicted score is within one point of the teacher's mark). They are defined as:

$$\text{exact} = \frac{1}{N}\sum_{n} \mathbf{1}[\hat{r}_n = r_n], \qquad \text{adj} = \frac{1}{N}\sum_{n} \mathbf{1}[\,|\hat{r}_n - r_n| \le 1\,], \tag{3.7}$$

where r_n and $\hat{r}_n$ are the true and predicted raw integer scores in points.

3.6.2 Explainability Protocol

Our explanation is *word attribution*: we highlight which words in the student's answer drove the predicted grade. Following the explainable-scoring approach of Kumar and Boulanger [13], the engine is *SHAP* (SHapley Additive exPlanations) [12]. Each word receives a Shapley value. This is the average change in the predicted score when that word is added to different groups of the other words. The Shapley value of word i is defined as:

$$\phi_i \;=\; \sum_{S \subseteq W \setminus \{i\}} \frac{|S|!\,(|W| - |S| - 1)!}{|W|!}\,\big[\,s(S \cup \{i\}) - s(S)\,\big], \tag{3.8}$$

Here, W is the set of answer words, S is a group of words kept in the answer, and $s(\cdot)$ is the score predicted using only those words. A positive ϕ_i means that the word increased the grade. We estimate Shapley values by testing different word orders with a permutation explainer, using a Monte-Carlo permutation method as a fallback. The reference answer

stays fixed. SHAP is the field-standard attribution and is *model-agnostic*: it calls only $s(\cdot)$, so it works for every pillar including the non-differentiable classical SVR, with no gradients or model internals. We also rank words across answers to show their *global* importance. We use each word's mean absolute SHAP value, following Kumar's view of global importance.

We judge an explanation by its *plausibility* [6]: whether the words it highlights look right to a teacher. A teacher grades a short answer by checking whether it carries the key content of the reference answer, so a plausible explanation should highlight answer words that also appear in the reference. We therefore measure plausibility as the fraction of the top-k SHAP words ($k = 20\,\%$ of the answer words) that occur in the reference answer; it lies in $[0, 1]$ and can be calculated for every model family.

CHAPTER 4
EXPERIMENT

4.1 Environment Setup

We implement the pipeline in Python. The classical and recurrent models run on a CPU; the Transformer encoders and the QLoRA-fine-tuned LLM need a CUDA GPU, which we run on the MISTI-HPC cluster (an NVIDIA A40) in `tmux` sessions. We use the `khmer-nltk` segmenter for Khmer word segmentation. We save the settings, per-answer predictions and metrics for every run. This allows us to calculate each reported number again without re-training. We use a fixed random seed (42) throughout.

4.2 The Experiment Grid

The study uses a list of experiments. Each tests one design choice so that we can link changes in the result to that choice. The grid contains twelve experiments. The first eight (`v01`–`v08`) build up the four model pillars from the cheapest classical baseline to the fine-tuned LLM; the last four (`exp09`–`exp12`) are the analyses that turn the trained models into the report's findings: the explainability study, the champion point-metrics, the data-cleaning ablation, and the classical hyperparameter sweep.

The study uses the `full` and `no10c` datasets. It tests cleaned and segmented text, with and without the question, together with each model's settings. We select each champion using the validation set rather than choosing one text format by hand (Section 3.3.1). The baseline (`v01`) and the BiLSTM grid (`v05`) are 12 configurations each; the Transformer grid (`v06`) is 36 configurations (three backbones $\times$ dual/cross $\times$ the text configurations). The full grid therefore contains well over a hundred configurations. The experiments use different hardware: the classical and BiLSTM families run on a CPU, while the Transformer and LLM grids run on the A40 GPU.

4.3 Before Training Phase

Before training, we fix the data and the experiment plan above. The experiments run in order from the cheapest classical baseline (`v01`) up to the fine-tuned LLM (`v08`), each building on and measured against the ones before it, The ensemble uses only classical, BiLSTM and Transformer configurations. The LLM is kept separate so that its contribution does not become hidden in the other results. Each experiment writes its own leaderboard and never edits another's output.

Table 9: The experiment registry. Each experiment isolates one design choice and writes its own leaderboard in the shared schema. "HPC" marks the GPU-only experiments.

ID	Pillar / role	What it isolates
`v01`	Classical	TF-IDF cosine and RBF-SVR baseline (12 cells per dataset).
`v02`	Classical	Post-hoc threshold calibration: refit the four class cut-points on validation.
`v03`	Classical	Add the question max-score as an SVR input feature.
`v03b`	RNN/Transf.	The max-score feature for the neural heads (BiLSTM CPU, Transformer HPC).
`v04`	Classical	A separate TF-IDF+SVR per max-score bucket, routed at test time.
`v05`	RNN	Full character BiLSTM + attention grid (12 cells per dataset). HPC.
`v06`	Transformer	Full mBERT / XLM-R / GTE dual and cross encoder grid (36 cells). HPC.
`v07`	Ensemble	Weighted mean of the top-3 cells across `v01`–`v06`.
`v08`	LLM	QLoRA fine-tuning of the KhmerGrader family, plus a zero-shot base. HPC.
`exp09`	Explainability	SHAP word attribution + plausibility for all four pillars.
`exp10`	Deployment	Champion point metrics: exact, adjacent, and raw-score error.
`exp11`	Data quality	The data-cleaning ablation across the two dataset variants.
`exp12`	Tuning	The classical hyperparameter sweep (C and TF-IDF max features).

4.4 Training Phase

For every cell, we train on the training split and select on the *validation* split, then evaluate the chosen configuration *once* on the held-out test split. We do not use test results to choose the model.

For the BiLSTM, GTE encoder and QLoRA fine-tuned LLM, we use *early stopping* based on validation QWK. We evaluate the validation set after each epoch and keep the saved model from the epoch with the highest validation QWK, rather than simply using the final epoch. In practice, validation performance stops improving after only a few epochs. Where both are recorded, training and validation scores stay close. Training for longer adds little, and these curves show no sign of over-fitting on the small dataset. The classical pillar has no such schedule: it trains in a single fit, with no epochs. This same validation-QWK selection rule is applied to every trainable cell of the grid.

We select hyperparameters by validation QWK. For the classical champion, which trains fast enough to sweep on a CPU, we run a grid over the SVR penalty $C \in \{0.1, 1, 10\}$ and the TF-IDF maximum features $\{5, 15, 30\} \times 10^3$; the validation-selected configuration changes the *test* QWK by less than 0.002 versus the defaults (Figure 5), so the classical

result changes little with these settings. For the neural, Transformer, and LLM pillars, we run the matching learning-rate sweeps (BiLSTM $\in \{5\text{e}{-4}, 1\text{e}{-3}, 2\text{e}{-3}\}$; Transformers $\in \{1\text{e}{-5}, 2\text{e}{-5}, 5\text{e}{-5}\}$; LLM QLoRA $\in \{1\text{e}{-4}, 2\text{e}{-4}, 3\text{e}{-4}\}$) with the same selection rule on GPU hardware.

Figure 5 shows this comparison: it places the test QWK of the default configuration next to the test QWK of the validation-tuned configuration, on the `no10c` data and, as a reference check, on the unfiltered corpus. The two bars are almost the same height in both cases. Changing the classical model's penalty and feature count changes test QWK by at most a few thousandths. The result therefore does not depend on one unusually favourable choice of settings.

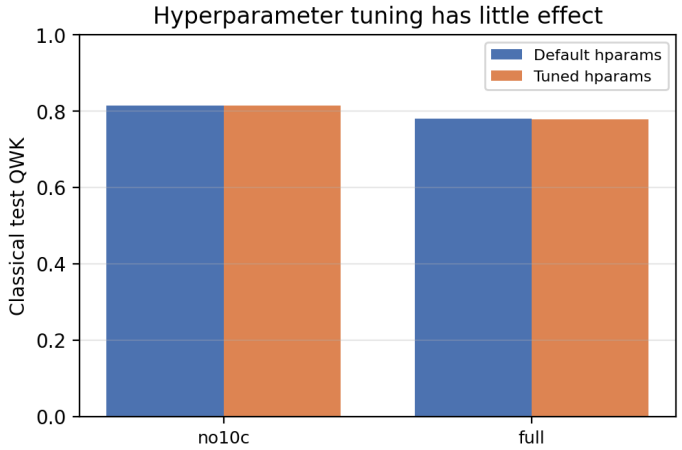


Figure 5: Classical hyperparameter sweep: default versus validation-tuned test QWK on the `no10c` data and, as a reference, on the unfiltered corpus. The similar bars show that changing penalty C and the TF-IDF feature count has little effect on the result.

4.5 Testing Phase

In the testing phase we compute every metric from the saved test-set predictions (Section 3.6). Each cell writes a self-contained run directory recording its exact configuration (the model, preprocessing mode, input format, and hyperparameters), the train, validation, and test metrics with the selected epoch, and the per-answer true and predicted score, label, and raw mark for every split. Each experiment also adds its results to a shared leaderboard. This puts results from different experiments in the same format for comparison.

Because every number comes from these saved predictions, we can reproduce every result in Chapter 5 without re-training: one script rebuilds the full set of figures from the saved predictions, with no GPU or network needed. The repository supplies source code, configuration and experiment scripts. The student corpus, predictions and model resources remain subject to their own access and distribution conditions. Making the source code

available does not mean that these research files are public. Repeating a calculation or training run requires approved access to the matching data, compatible model resources and recorded settings.

4.5.1 The Live Prototype

We provide the explanation feature through a working web application. The interaction mirrors what a teacher already does:

1. the teacher types (or pastes) the **question**, the **reference answer**, and the **student's answer**;
2. the teacher **chooses a model** from the four pillars;
3. the system returns a **score**, a **word-attribution explanation** (a text-highlight heatmap), and **written feedback**.

The application uses the same SHAP word attribution evaluated in the explainability study. The explanation is therefore a main feature of the prototype. The heatmap and feedback always use the readable segmented tokens (Section 3.3). The tool is designed to *assist a teacher*. The teacher stays involved in the decision and keeps responsibility for the final grade.

The written feedback comes from an open-source LLM served over an OpenAI-compatible endpoint, with a rule-based fallback when no model server is reachable, so the tool never depends on a proprietary grading service. SHAP scores each answer many times, and the LLM must generate a new output each time. We therefore provide an on/off *toggle* for the explanation. A teacher can turn it off to get a score faster, or turn it on to see the word-attribution heatmap. The application runs locally and is not publicly hosted.

The proposed mark and explanation support the teacher's review. The teacher checks the answer against the question and reference and retains responsibility for the final mark. Agreement scores and reference-overlap highlights do not replace the teacher's judgement. The prototype and its feedback have not yet been evaluated with teachers in a classroom field study.

CHAPTER 5
RESULTS AND DISCUSSION

How to read the numbers. Two numbers matter most: **QWK** (how well a model's grades agree with the teacher overall) and **exact match** (how often it gives the same mark out of N). All headline values are *uncalibrated* and reported on the `no10c` test set, except that the encoder champion was selected using the dataset before 10C was removed. Its QWK therefore gives a useful comparison, but it is not a direct comparison on exactly the same test set.

5.1 Results

5.1.1 Per-Pillar Accuracy (RQ1)

The fine-tuned LLM performs best on the reported grading measures. It has the highest Quadratic Weighted Kappa *and* the best deployment scores. The three smaller model families have similar, lower scores. Table 10 lists the results, and Figures 6–10 display them.

Table 10: Per-pillar champion results on the test set (uncalibrated, random split; macro-averaged F1). The fine-tuned LLM leads on every metric; the three smaller pillars cluster lower. The final row is the untrained Qwen base graded *zero-shot*, a reference that shows the fine-tuning lift, not a fifth pillar. QWK, accuracy, and macro-F1 are the champion-selection metrics; the exact integer-mark column is read from each champion's per-configuration leaderboard.

Pillar	QWK	Accuracy	macro-F1	Exact	Adjacent (±1)
Classical (SVR)	0.802	0.65	0.47	0.28	0.72
RNN (BiLSTM)	0.789	0.69	0.53	0.34	0.70
Transformer (GTE)	0.777	0.68	0.51	0.49	0.66
LLM (Qwen-KhmerGrader, FT)	**0.850**	**0.81**	**0.80**	**0.67**	**0.80**
Qwen base, zero-shot (ref.)	0.529	0.42	0.30	0.25	0.43

Reading Table 10 model by model, ranked by QWK:

- The **LLM (Qwen-KhmerGrader)** reaches the highest QWK at **0.850** and leads on *every* other metric as well: accuracy 0.81, macro-F1 0.80, and adjacent agreement 0.80. It performs best on both the research measure and the measures used to assess classroom deployment.
- The **Classical (SVR)** is the strongest of the smaller pillars at QWK **0.802**, a strong result for a model that trains in about 30 seconds on a CPU. Its macro-F1 (0.47), however, is

the lowest of the four.

- The **RNN (BiLSTM)** follows at QWK **0.789**, with the best accuracy (0.69) among the three smaller pillars.
- The **Transformer (GTE)** sits at QWK **0.777**; the encoder is evaluated on the corpus before the 10C removal, so its figure is indicative rather than a strict head-to-head.

The fine-tuned LLM has a clear lead. The three smaller model families have QWK scores in a narrow 0.78–0.80 range. KhmerGrader scores roughly 0.05–0.07 higher, with an even larger lead on the deployment measures. Fine-tuning explains this improvement: the same Qwen base scores only **0.529** QWK zero-shot (Table 10, last row). Fine-tuning adds **+0.32** QWK. Figure 6 shows this directly: the vertical axis is test QWK and each bar is one pillar, and the LLM bar stands above the cluster of the other three.

Key insight (RQ1). The fine-tuned LLM (KhmerGrader) is the best pillar on the primary research metric *and* on deployment, QWK 0.85 versus 0.78–0.80 for the classical, RNN, and encoder models, and a decisive lead on exact-mark agreement. The gain comes from fine-tuning on real Khmer data, rather than model size alone. The same base reaches only 0.53 zero-shot.

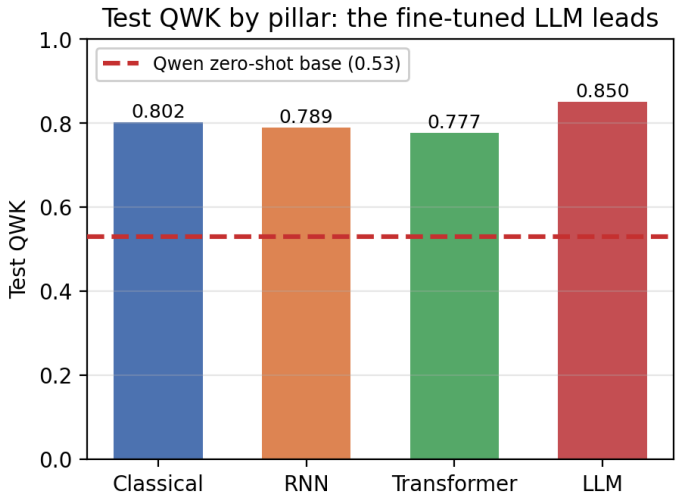


Figure 6: Test QWK of the four pillars. The fine-tuned LLM (KhmerGrader) bar stands clearly above the cluster of the three smaller pillars, which fall in a 0.78–0.80 band; the red dashed line marks the untrained Qwen zero-shot base for reference.

The table also shows two broader patterns.

First, macro-F1 is lower than accuracy for every model except the LLM (classical 0.47 versus 0.65). Full-credit answers are common in the data, and the cheaper models perform poorly on less common grades. Figure 7 shows this pattern through macro-averaged precision, recall and F1 for each model family. "Macro" means every grade has equal

weight, even when it is rare. The classical, RNN and Transformer models score near 0.45 to 0.55 on all three measures. The LLM reaches about 0.80 on each. It is the only model that handles rare partial-credit grades well, rather than mainly predicting the common grades.

Second, Figure 8 shows the training and test results so that over-fitting is visible. The classical model has the largest gap (QWK $0.91 \rightarrow 0.80$, accuracy $0.79 \rightarrow 0.66$). The deep models have smaller gaps, with the LLM at $0.91 \rightarrow 0.84$ QWK and $0.86 \rightarrow 0.84$ accuracy. The BiLSTM even scores slightly higher on test than on training data. We use validation QWK and early stopping to select models, then report their test results. We favour a smaller training-to-test gap over a high training score that does not carry over to test data. Later in this section, we also compare these gaps with the Arabic reference study.

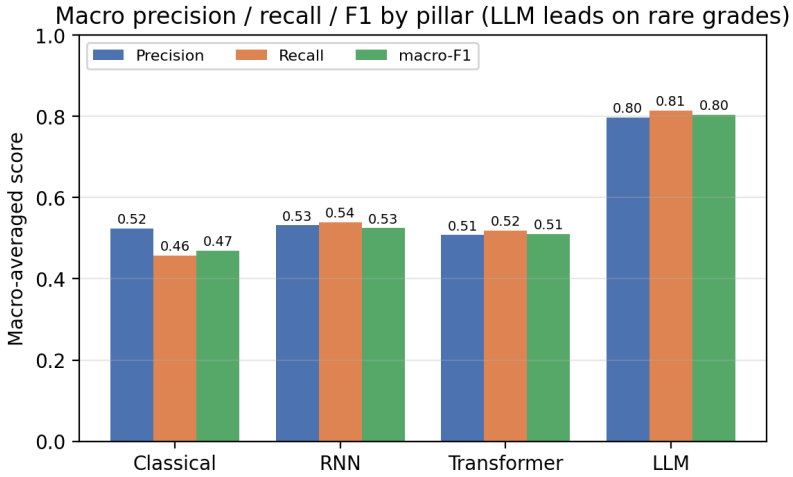


Figure 7: Macro-averaged precision, recall, and F1 of each pillar. Macro-averaging weights every grade equally, so the LLM's jump to about 0.80 on all three shows it is the only pillar that scores the rare partial-credit grades well.

The confusion matrices (Figure 9) and predicted-versus-true scatter plots (Figure 10) show the same pattern in two ways. Each confusion matrix places the true grade on the vertical axis and the predicted grade on the horizontal axis. Each cell counts the answers with that pair of grades. A perfect model would place every answer on the diagonal. All models have many answers near this line, but the classical model has the most disagreements and the LLM has the fewest. The scatter plots compare raw marks. Each point is an answer, with its true mark on the horizontal axis and predicted mark on the vertical axis. The dashed line marks perfect agreement. The LLM's points are closest to the line. The classical model's points are most spread out, especially because it predicts marks nearer the middle for answers with high true marks.

5.1.2 Research versus Deployment (RQ2)

The fine-tuned LLM leads on QWK. Its lead is *larger* on the deployment measures that matter to a teacher:

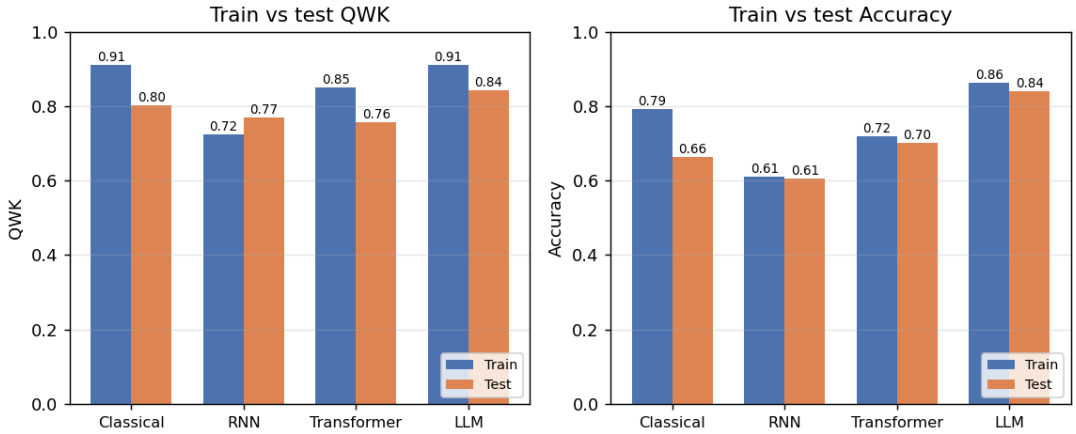


Figure 8: Train versus test QWK (left) and accuracy (right) for each pillar's champion (values from the per-configuration leaderboard run). The classical model has the widest train-to-test gap, while the deep pillars stay tight and the BiLSTM even scores slightly higher on test than train, so the reported test numbers are not inflated by over-fitting.

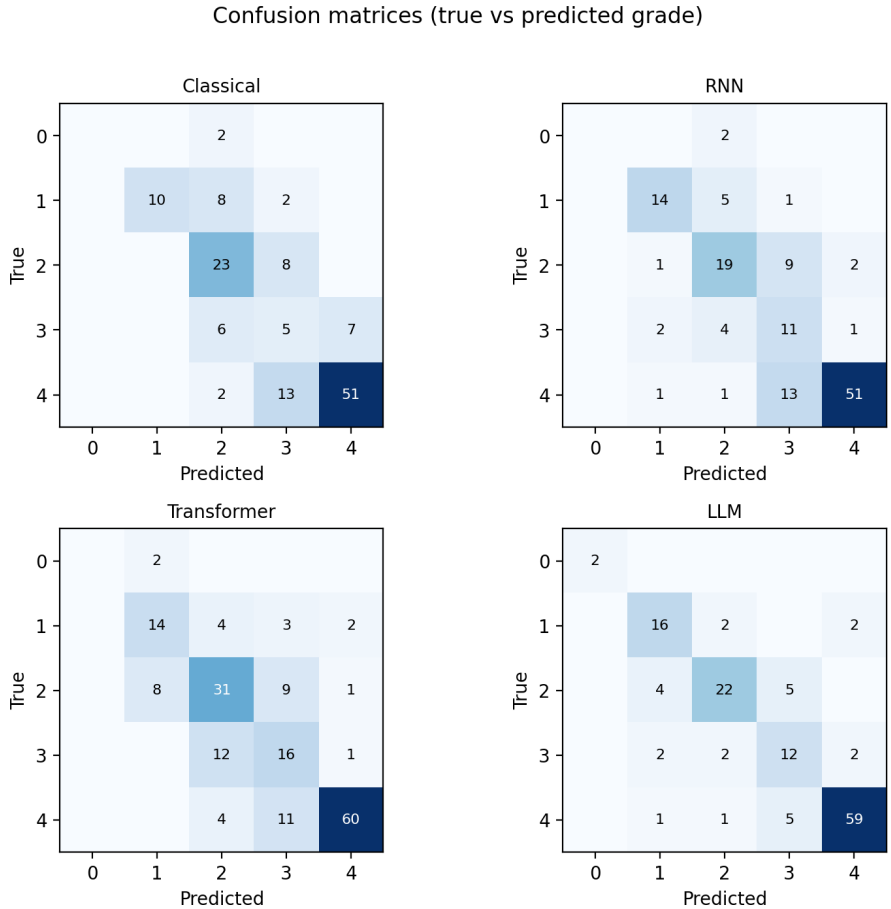


Figure 9: Confusion matrices (true grade on the vertical axis, predicted grade on the horizontal axis; cell counts). Counts on the diagonal show agreement. Counts away from it show disagreements, which are most common for the classical model and least common for the LLM.

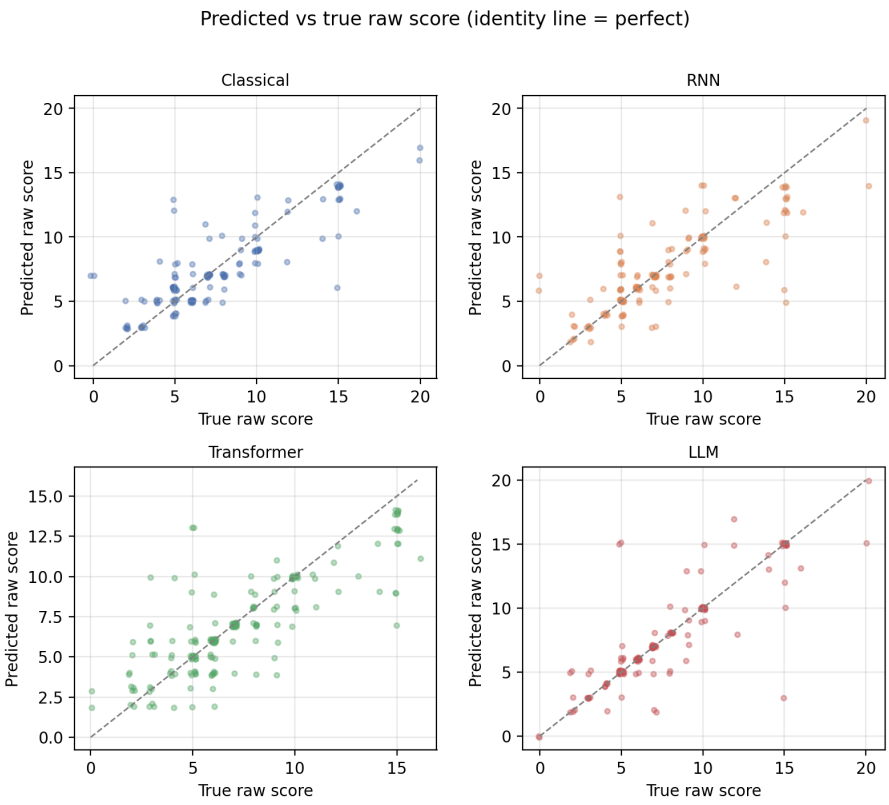


Figure 10: Predicted versus true raw score in points (one point per answer; the dashed line marks perfect agreement). The LLM's points lie closest to the line; the classical model predicts high marks nearer the middle.

- **Exact mark:** LLM **0.67**, ahead of the three smaller pillars.
- **Within one point:** LLM **0.80**.
- **Lowest average point error** of any pillar.

Figure 11 shows the deployment gap. For each pillar it places two bars, the fraction of test answers scored *exactly* right and the fraction scored within one point, both on the same 0-to-1 vertical axis. Every model has a high within-one-point bar, but the exact-match bars differ greatly. The LLM reaches 0.67, while the cheaper models score much lower. These bars show how often each model gives exactly the recorded teacher mark.

Figure 12 explains *why* the LLM wins here. It compares two histograms for the same test answers: the teacher's marks and the LLM's predicted marks. The two distributions have the same shape, including the high number of full-credit marks. The predictions do not gather mainly in the middle. The cheaper regression models minimise squared error and tend to predict the common middle grades. The LLM better matches the full range of teacher marks. The choice of tool therefore depends on the goal: the low-cost classical model with SHAP for a quick check, or the fine-tuned LLM when matching the exact mark matters.

Key insight (RQ2). The fine-tuned LLM leads on QWK and *widens* that lead on exact-mark agreement: it is the only pillar that reproduces the teacher's exact mark often enough to hand back unchanged. The models differ most on deployment quality. Research agreement and classroom readiness should therefore be reported separately.

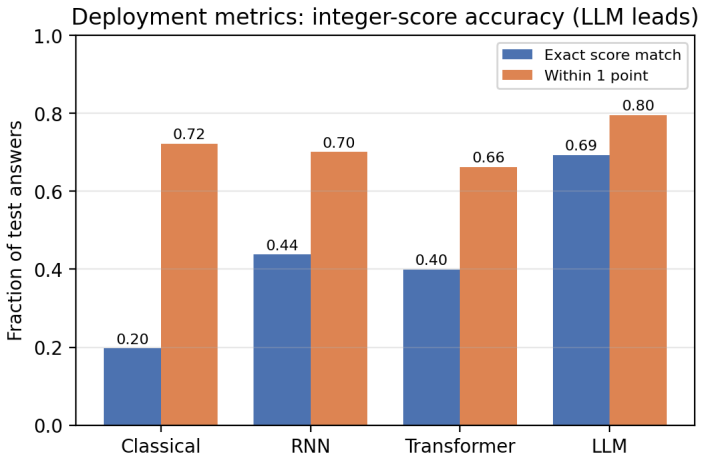


Figure 11: Deployment metrics: the fraction of test answers scored exactly right and within one point. Within-one-point is high for all, but exact match separates the pillars, with the LLM clearly ahead.

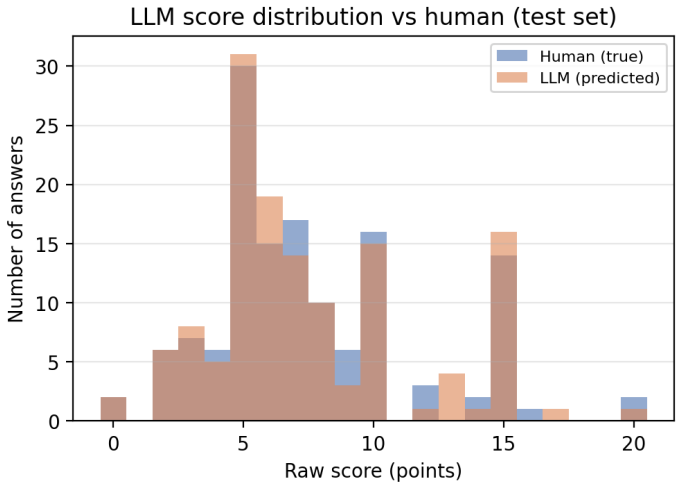


Figure 12: Distribution of human marks and LLM-predicted marks on the same test answers (points axis). The predicted distribution matches the human one, including the full-credit spike, rather than gathering around the average.

Efficiency and real-time scoring. A tool that assists teachers also needs to respond quickly. Table 11 therefore reports the deployment *cost* for each family: the time to train once, and the time to return a score *together with* its SHAP explanation for one answer. The classical model is fastest, about two seconds per explained answer on a CPU, while the LLM is the slowest at over a minute, because each SHAP evaluation is a full generation; this is

why the prototype offers the explanation as a toggle (Section 4.5.1). For every family, a score without an explanation is much faster. Calculating the explanation takes almost all of the response time shown.

Table 11: Per-pillar efficiency: one-off training time and the per-answer latency to return a score *with* its SHAP explanation (the explainable-mode cost a teacher experiences), measured on the test set under the capped SHAP budget. Classical and BiLSTM run on CPU; the encoder and LLM on GPU.

Pillar	Training time	Per-answer (score + SHAP)
Classical (SVR)	∼30 s (CPU)	∼2 s
RNN (BiLSTM)	∼150 s (CPU)	∼13 s
Transformer (GTE)	∼140 s (GPU)	∼10 s
LLM (Qwen-KhmerGrader)	∼3.8 h (GPU)	∼76 s

5.1.3 The LLM Pillar: Zero-Shot Bases vs the KhmerGrader Family

The LLM pillar is the KhmerGrader family. We fine-tune three open base models with the same QLoRA method to test how much the choice of base model matters. We compare each with its own zero-shot base on the same 909 test set (Table 12, Figure 13). This follows recent work comparing fine-tuned and prompted LLMs for grading [15, 19] and fine-tuning open models such as Qwen for scoring [16].

Table 12: KhmerGrader family: fine-tuned test QWK for each base (909 test set). The Qwen base scores 0.529 QWK *zero-shot* (Table 10), a $+0.32$ fine-tuning lift. The Qwen QWK is from `champion_metrics.csv`; the Gemma and SEA-LION QWK are those plotted in Figure 13 (from the exp08 fine-tune run).

KhmerGrader model	Base size	Fine-tuned QWK
Qwen-KhmerGrader-4B	4B	**0.850**
Gemma-KhmerGrader-4B	4B	0.844
SEA-LION-KhmerGrader-E2B	2.3B	0.713

Reading Table 12 model by model, ranked by fine-tuned QWK:

- **Qwen-KhmerGrader-4B** is the QWK champion at **0.850**, lifting its 0.529 zero-shot base by $+0.32$.
- **Gemma-KhmerGrader-4B** follows closely at QWK **0.844**, within 0.01 of the champion.
- **SEA-LION-KhmerGrader-E2B**, on the smallest base (about 2.3B), reaches QWK **0.713**.

All three bases are instruction-tuned, so each can follow the grading prompt without worked examples. QLoRA fine-tuning improves all of them well above their zero-shot scores. This shows the importance of the fine-tuning data compared with the base model's existing Khmer ability. The two 4B bases score within 0.01 QWK of each other. The 2.3B base scores lower. This suggests that model size helps, while learning from the grading data has the larger effect. Each base is fine-tuned on the `no10c` training split; the champion prompt includes the question, which only the generative pillar can exploit, consistent with the input-format ablation (Section 5.1.4).

Figure 13 shows the improvement. For each KhmerGrader model, it places the zero-shot base QWK next to the fine-tuned QWK. The difference is the gain from QLoRA. The three bases start from similar zero-shot levels (about 0.51 to 0.53) and all rise sharply after fine-tuning: the two 4B models reach the highest scores (0.85 and 0.84), while the 2.3B model scores lower (0.71). Fine-tuning provides most of the improvement, with base model size making a smaller contribution.

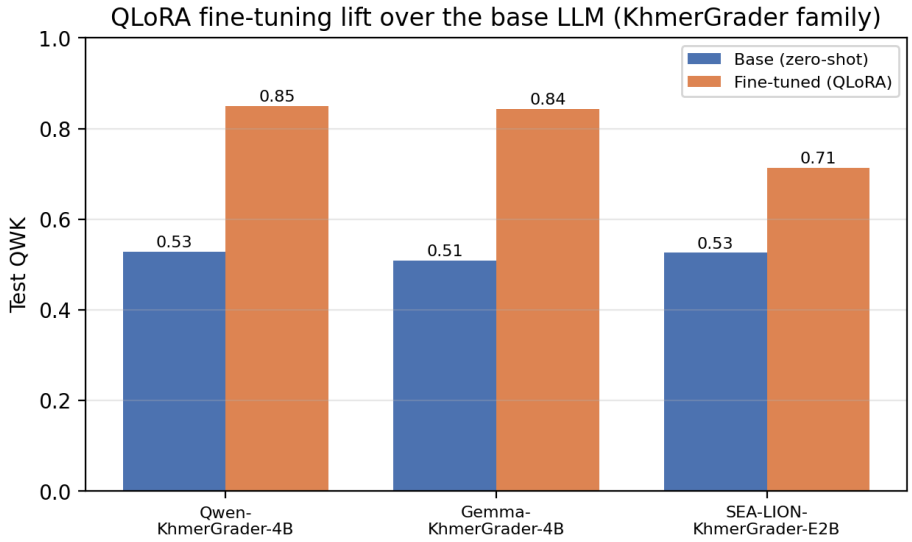


Figure 13: Zero-shot base versus QLoRA-fine-tuned test QWK for each KhmerGrader model. Bases start at very different heights and all rise sharply after fine-tuning; the QLoRA lift, not the base's prior Khmer ability, drives the result.

Specialist versus frontier models. A natural question is whether a 4B fine-tuned KhmerGrader can match much larger frontier models. Table 13 compares the KhmerGrader champion against four frontier LLMs prompted zero-shot on the same `no10c` test set (Section 3.4.5), each in both the bare-integer and reasoning-enabled settings, on five metrics. The per-model unparsed-response rate is reported in Appendix E. All frontier numbers are a dated API snapshot; the fine-tuned KhmerGrader, code, and data remain reproducible.

The 4B fine-tuned KhmerGrader (QWK 0.850) scores higher than *every* frontier

Table 13: Fine-tuned KhmerGrader champion vs. frontier LLMs prompted zero-shot on the `no10c` test set (137 answers; frontier metrics use each model's parsed subset, Appendix E), each in both the *bare* (integer answer) and *reasoning*-enabled settings. Frontier rows are a version-pinned API snapshot evaluated 2026-06-24; pinned versions: GPT `gpt-5.5-20260423`, Claude `claude-opus-4.8`, Gemini `gemini-3.5-flash-20260519`, DeepSeek `deepseek-v4-flash-20260423`. Metrics: QWK, accuracy (Acc), macro-F1, exact integer-mark, and within-one (± 1).

Model	Setting	QWK	Acc	F1	Exact	± 1
Qwen-KhmerGrader-4B	**fine-tuned**	**0.850**	**0.81**	**0.80**	**0.67**	**0.80**
GPT-5.5	bare	0.684	0.59	0.53	0.48	0.67
	reasoning	0.708	0.58	0.50	0.45	0.65
Claude Opus 4.8	bare	0.662	0.54	0.37	0.35	0.63
	reasoning	0.734	0.62	0.61	0.43	0.69
Gemini 3.5 Flash	bare	0.555	0.60	0.45	0.46	0.61
	reasoning	0.585	0.58	0.42	0.45	0.68
DeepSeek V4	bare	0.553	0.48	0.38	0.34	0.52
	reasoning	0.579	0.49	0.37	0.31	0.56

model in this benchmark. It leads the best frontier model (Claude Opus 4.8, QWK 0.734) by more than 0.11. The gap is larger for the other three. This holds even when the frontier models are allowed to reason before answering (the per-setting breakdown is in Appendix E). For this task with limited data, a small specialist fine-tuned on a few hundred real Khmer answers performs better than much larger proprietary models used zero-shot. This is the main practical reason for the KhmerGrader approach.

Key insight. For this benchmark, fine-tuning a small open model on Khmer grading data helps more than the larger size and general ability of a frontier LLM: the 4B KhmerGrader leads the best frontier model by over 0.11 QWK, reasoning enabled or not.

5.1.4 Design-Choice Ablations

We study each design choice separately while keeping the other factors fixed. The full leaderboards for the tested combinations are supplied with the code. The choice of *which data* is kept is validated in Section 3.2.3: removing the inconsistently graded 10C subset raises the classical champion's QWK by about $+0.044$, while a cleaning refinement moves it by at most 0.003. The remaining factors are checked here.

Text representation. Between the cleaned and the segmented stage of the pipeline (Section 3.3.1), each family's champion selects the representation that suits it: the classical char-n-gram model wins with the *segmented* text, because explicit word boundaries help a

bag-of-n-grams, while the BiLSTM, encoder, and LLM win with the *cleaned* text and do not need word segmentation, since they learn their own units from the text stream itself.

Input format. Between the (answer, reference) pair and the question-prefixed input, the surface models (classical, BiLSTM, encoder champions) win *without* the question: the question text is the same for all answers to that question, so it adds no word-level information that helps these models distinguish the answers. The LLM benefits from the question because it can actually read and use it.

Max-score feature. Adding the question's normalised maximum score improves deployment results at very little extra cost (about $+4.5$ points of exact match for the neural heads on the 909 set). It directly tells the model the question's scale, instead of adjusting grade boundaries using a small validation set.

Calibration. Adjusting grade boundaries after training helps some models but hurts others, even with the same validation-based rule. Its benefit is therefore uncertain and depends on the model. The boundaries use only $\sim$134 validation rows, so improvements do not reliably carry over to test data. We report calibration as a separate experiment. The headline results (classical QWK 0.802, BiLSTM 0.789) are uncalibrated.

Hyperparameters. The classical champion's test QWK moves by less than 0.002 across the SVR penalty and TF-IDF feature-count sweep (Chapter 4, Figure 5), so the result does not depend on a lucky hyperparameter choice.

Key insight. On this dataset, the data selection in Section 3.2.3 affects results as much as the model design. The low-cost max-score feature also gives reliable gains. The other factors (text representation, input format, calibration and hyperparameters) have smaller effects or depend on the model.

5.1.5 Explainability (RQ3)

The word highlights point at the content a teacher looks for, and they do so for every model. The method is *SHAP* word attribution (Section 3.6.2), following the explainable-scoring work of Kumar and Boulanger [13]. SHAP is model-agnostic, so the same method explains all four pillars, including the non-differentiable classical SVR. Following Kumar, we report three things: a *local* explanation for each answer (the heatmap), a *global* ranking of important words across answers, and a plausibility check that compares highlighted words with the reference answer. The classical, BiLSTM, and encoder pillars run a full SHAP sweep; for the LLM, where each SHAP evaluation is a full generation, we

run SHAP under a *capped* evaluation budget so plausibility is still computed for every pillar, with illustrative LLM heatmaps in Appendix F.

Table 14: SHAP plausibility of each model: the fraction of the top-20% SHAP words that also appear in the reference answer (higher means the highlighted words are rubric-relevant). The classical and BiLSTM pillars are scored on $n = 66$ test answers; the encoder and LLM on a capped $n = 18$, since each LLM SHAP evaluation is a full generation.

Model	SHAP plausibility	n
Classical (SVR)	0.63	66
RNN (BiLSTM)	0.60	66
Transformer (GTE)	**0.68**	18
LLM (Qwen, capped)	0.61	18

Reading Table 14, all four pillars land in a similar plausibility band, roughly 0.60 to 0.68: the encoder is highest (0.68) and the BiLSTM lowest (0.60), with the classical and LLM pillars in between (0.63 and 0.61). The encoder and LLM use a smaller, capped sample, so these close scores should not be treated as a firm ranking. The main point is that SHAP gives a *plausible* explanation for every model family. Choosing a different model does not mean losing this kind of explanation. For every model, the top SHAP words mostly carry subject meaning rather than only serving a grammatical role. This makes the highlights a possible basis for teacher feedback.

Globally important words. Combining SHAP values across the explained answers gives a global ranking of the words a model uses most (Kumar's global-importance view). For the classical model the highest-ranked words are mostly subject-content terms, for example សេដ្ឋកិច្ច (economy), ការនេសាទ (fishing), ប្រទេស (country), ប្រជាជន (people), and កម្ពុជា (Cambodia), confirming that the model grades largely on the content of the answer rather than on surface form. Figure 14 shows this global-importance view for all four pillars as the standard SHAP bar summary: each bar is one answer word, with length the mean absolute SHAP value across the explained answers.

The local SHAP heatmap shades each readable word of a real Khmer answer by its SHAP value. Darker shading means a larger contribution to the predicted grade. The reader can quickly see which words influenced the score, rather than seeing only a number. The prototype can use these same attributions to help give feedback. Example SHAP heatmaps on real answers for each pillar appear in Appendix F.

Key insight (RQ3). One model-agnostic method, SHAP, gives a plausible word-level explanation for every pillar, including the non-differentiable classical model, so explainability is not traded against accuracy: the cheapest and the most expensive pillars are explained the same way.

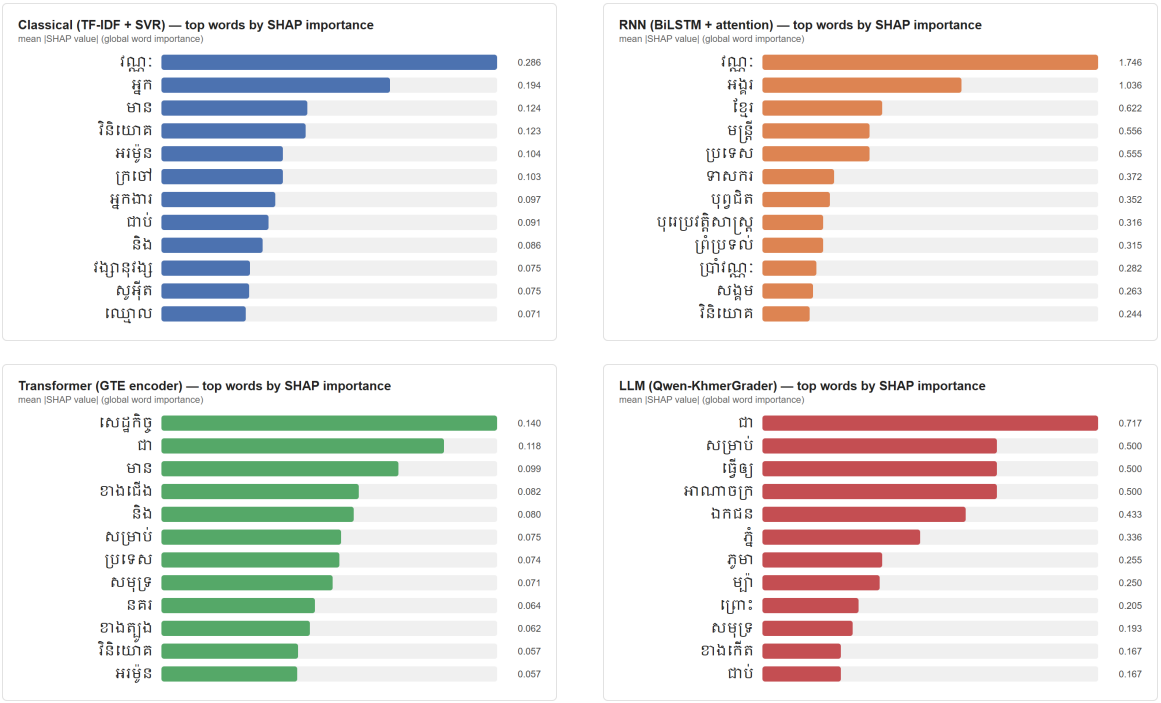


Figure 14: Global SHAP word importance for each pillar: the top answer words ranked by mean absolute SHAP value (the standard SHAP global-importance bar plot). The top words are mostly content words, so each pillar grades on answer content rather than surface form.



Figure 15: Sample of a Khmer word-attribution heatmap (classical model, SHAP).

Interpreting the reference-overlap proxy. The highlights help a teacher inspect which parts of the student's answer influence the predicted mark. Plausibility checks word overlap with a reference taken from a student answer. That reference may be incomplete. The measure therefore estimates content relevance; it does not prove that the explanation covers every correct point an answer could make. Classical and BiLSTM use 66 answers each, while encoder and LLM use 18 each; the displayed order is not a ranking based on the same sample. We still need an evaluation with teachers to find out whether they understand and find these explanations useful.

5.2 Discussion

Taking the models in turn, the results follow from how each one represents a Khmer answer.

The **Classical** model (TF-IDF with RBF-SVR) is the strongest of the smaller pillars (QWK 0.802) because character n-gram overlap is a strong signal for short factual answers, and the support-vector regressor is hard to over-fit on a small corpus. This focus on word overlap also explains its weaker exact-mark score and macro-F1 (0.47). Its predictions tend toward common grades, and it misses the small differences in partial credit that a teacher recognises.

The **RNN (BiLSTM)** reaches QWK 0.789 because additive attention over the BiLSTM captures order and local context that bag-of-n-grams misses, while staying small enough to fit the roughly 600 training rows without over-fitting. It still trails the LLM on the exact mark, as it predicts a single normalised score rather than reasoning about the rubric.

The **Transformer (GTE)** sits at QWK 0.777 because the multilingual encoder brings useful features based on context but little Khmer pre-training, so it does not beat the much cheaper pillars on this corpus; recall that it was selected on the corpus before the 10C removal, so its figure is indicative rather than a strict head-to-head.

The **LLM (Qwen-KhmerGrader)** is the highest-performing pillar on the reported grading metrics (QWK 0.850, exact 0.67, adjacent 0.80) because QLoRA fine-tuning lets a generative model learn the scoring rubric directly from the question, reference, and answer, producing marks that follow the human distribution instead of collapsing toward the mean. It leads the three smaller pillars on overall agreement *and* widens that lead on the deployment metrics.

Choosing the "best model" requires considering all three areas. The fine-tuned LLM leads on research agreement (QWK) *and* deployment (the exact mark). SHAP gives similarly plausible word-level explanations for *every* model, in a narrow 0.60 to 0.68 range. Lower-cost models with SHAP therefore remain useful when computing resources or the ability to inspect a model are the main concern.

We can also place the numbers against the Arabic anchor [18] in two careful ways. First, on the *same* unweighted Cohen's κ, our models score 0.49 (classical), 0.55–0.57 (encoder and RNN), and 0.72 (LLM), versus their 0.48 (BERT) and 0.60 (best transformer); these numbers provide context, but cannot rank the studies because they use different datasets. Second, and more fairly, the *generalisation gap*, or difference between training and test performance: their best transformer over-fits by $+18.5$ points (95.67/77.22 accuracy), whereas our deep cells are far tighter ($+1.4$ for the GTE encoder; $+3.0$ for Qwen, 83.3/80.3), and even the classical model ($+15.6$, 78.6/63.0) stays under their transformer. These

comparisons make the training-to-test differences visible.

For a classroom, the implication is concrete: a cheap classical model that explains itself with SHAP is an attractive default; the LLM is the right choice when the exact mark matters; and in both cases the explanation shown to a teacher should highlight words that are plausible, matching the reference content. The main limit is that every number measures agreement with a *single* teacher. We have not measured how closely different human graders agree: reported Khmer ASAG scores, including ours, should be read as agreement with one grader on a small, four-subject corpus.

CHAPTER 6
LIMITATIONS AND FUTURE WORK

6.1 Limitations

A clear benchmark needs to explain its limits. Below, we connect each limitation to a result so that the reader can understand the problem and how much it may affect the findings.

- **Student-derived reference answers.** Official answer keys were unavailable, so the highest-scoring student response was used for each question. These references may be incomplete or contain errors. This can affect both reference-based grading and the SHAP reference-overlap measure. References checked by subject experts would improve future evaluation.

- **Single grader, no inter-annotator agreement.** We could not include a second grader, so we do not know how closely human graders would agree. All results measure agreement *with this teacher* and may reflect that grader's biases. Removing the group with clear grading inconsistencies reduces part of this problem, but does not solve it. The 10C Biology case (Section 3.2.3) also shows that one grader does not always give consistent marks.

- **Small corpus and few questions.** With 909 curated answers (of 1,184 collected) across only 41 distinct questions, the dataset covers four subjects from two schools and does not represent the country. Some groups are very small: Earth Science in particular contributes only nine answers.

- **Partial cross-pillar comparability.** The classical, RNN, and LLM champions share the `no10c` test set, while the encoder champion was selected on the corpus before the 10C removal, so its metrics are computed on a slightly different test size. The encoder's QWK is therefore a useful reference rather than a direct comparison on the same test set. The LLM still leads the classical, RNN and encoder models on the reported research and deployment measures; comparisons using the shared `no10c` split are the most direct.

- **Explainability scope.** SHAP plausibility is reported for all four pillars on the `no10c` split; the LLM uses a *capped* SHAP evaluation budget, since each evaluation is a full generation, so its plausibility is an approximation of the full sweep. Plausibility is an automatic word-overlap measure. It does not compare highlights with reasons given by human teachers, so it only estimates how useful the explanations may be for education.

- **Frontier baselines are a dated API snapshot.** The frontier-model comparison (Section 3.4.5) uses closed services that are regularly updated. The same numbers may not be reproducible if a provider changes or removes a model version. To reduce this problem, we record each exact version and the evaluation date, and save the raw responses; KhmerGrader weights, code, and data remain fully reproducible.

- **LLM hallucination and output validity.** A generative grader can produce text instead of a valid score, a number outside the allowed range, or a confident but incorrect mark. We limit the first two problems by asking for a score only, extracting the first integer, and limiting it to $[0, \text{Max}]$. An output with an extractable integer is therefore converted to a mark within the allowed range (Section 3.4). If no score can be extracted, we report the response as *unparsed* rather than assigning a replacement score (Appendix E). This was rare for fine-tuned KhmerGrader but common for some zero-shot frontier models (Gemini returned no parseable score on 65 of 137 bare prompts). A valid-looking but *incorrect* score is still a risk. The agreement measures (QWK, exact) directly measure this problem. SHAP can help a teacher spot it by showing which words influenced the mark. The prototype's *written feedback* has the greatest risk of invented or incorrect statements. It uses SHAP attribution as a basis, and the application uses rule-based feedback when no language-model server can be reached. However, we have not yet formally checked the feedback's factual accuracy (see future work).

- **No formal field study.** The live prototype shows that the tool can work, but we have not yet tested its classroom value with teachers in a controlled study.

6.2 Future Work

The following steps would address these limitations.

- **A second grader.** A second grader would let us measure agreement between humans. Model results could then be compared with that measured agreement, rather than only with one grader.

- **Few-shot prompting of the bases and frontier models.** We prompt only zero-shot; adding worked examples (for instance one example per grade band) could improve the prompted baselines. The number and choice of examples matter. Models may favour the most common labels or the examples shown last. Future work should therefore test few-shot prompts carefully, including examples near grade boundaries.

- **A larger, multi-school corpus.** Collecting answers from more than the two schools and four subjects, with more different questions, would better represent Cambodian secondary education.

- **Khmer-specific pretraining.** Continued pretraining of a multilingual encoder on Khmer text, or a Khmer-native encoder, may close the gap left by the limited Khmer seen during generic multilingual pretraining.
- **A human plausibility study.** Teachers could mark the words they consider relevant to a grade. We could then compare SHAP highlights with the teachers' choices, instead of relying only on automatic reference overlap.
- **Cross-lingual transfer.** Transferring scoring ability from high-resource languages to Khmer (for example, via multilingual representations or cross-lingual contrastive learning) could help when very little training data is available.
- **Feedback quality.** The prototype's written feedback should be tested for usefulness in classroom learning, building on rubric-aligned feedback work.
- **A teacher field study.** A controlled study of how Cambodian teachers use, trust, and benefit from the prototype would show how well the explanations work in real use.

The highest priority is to add a second grader. This addresses a main limit on how widely the findings can apply: it would let us compare results with measured human agreement instead of one teacher alone. The next step is a larger dataset from more schools and with more different questions. The other improvements would build on these two steps.

CHAPTER 7
CONCLUSION

This study developed the first reproducible and explainable benchmark for automatic short-answer grading in Khmer. The full corpus contains 1,184 answers graded by a teacher and written by 203 students from two Cambodian high schools, one in a province and one in a city. It covers 41 questions from Biology, History, Geography, and Earth Science. We removed the 10C Biology group because its marks were inconsistent. The remaining 909 answers formed the cleaned dataset used in the main experiments. The study compared four model families in one grading pipeline: a classical model, a BiLSTM, multilingual Transformer encoders, and fine-tuned open-source large language models. It also applied SHAP word attribution across the four model families and developed a working prototype.

The results answer the three research questions. For RQ1, the fine-tuned Qwen-KhmerGrader-4B achieved the strongest agreement with the teacher, with a QWK of 0.850. It outperformed the classical model (0.802), the BiLSTM (0.789), and the Transformer encoder (0.777). The same Qwen base model achieved only 0.529 QWK without fine-tuning, showing that training on the Khmer grading data produced a gain of about 0.32 QWK. The fine-tuned model also performed better than the four frontier language models tested with zero-shot prompting. The strongest of these, Claude Opus 4.8, achieved a QWK of 0.734.

For RQ2, Qwen-KhmerGrader-4B also produced the best exact-score results. It matched the teacher's exact mark for 67% of the test answers and gave a mark within one point for 80%. These results make it a useful option to test when matching raw marks matters. They do not prove that it is ready for classroom use. The prototype and its feedback still need to be evaluated with teachers. However, the classical model remains useful when computing resources are limited. It achieved a QWK of 0.802 and can be trained in about 30 seconds on a CPU. The experiments also showed that removing data with inconsistent marks helped more than many changes to the model design. Including the maximum possible score as an input feature also improved results with little extra cost.

For RQ3, the same model-agnostic SHAP method produced word-level explanations for all four model families, including the classical support-vector regressor. Their plausibility scores, which measure overlap with the reference answer, ranged from approximately 0.60 to 0.68. The highlighted words were generally content words that also appeared in the reference answers. This allows a teacher to inspect which parts of an answer influenced a predicted score. These results give early evidence that the same explanation method works across different grading models. Future work still needs to test the explanations with teachers.

Overall, the results show that fine-tuning a relatively small open-source language model on data from the target subject and language can be more effective than using much larger general-purpose models without training. They also show that model quality should not be judged by QWK alone. Exact agreement, computing cost, explanation quality, and the needs of teachers should also be considered when selecting a model.

The main contribution of this study is a starting point for further research on Khmer educational NLP. The dataset design, source code, evaluation steps and prototype provide a basis for future comparisons and improvements. Using the private research files still requires approved access. At the same time, the results are based on a limited corpus and agreement with one teacher. Future studies should include more schools, subjects, students and independent graders. They should also test the prototype and its explanations in classrooms.

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CHAPTER A
APPENDICES

A Hyperparameter Inventory

Table 15 gathers the main training settings, or hyperparameters, in one place to help readers repeat each configuration. Classical and BiLSTM settings are the defaults used throughout; the Transformer and LLM settings follow standard fine-tuning practice with validation-selected learning rates (Section 4.4). The model input representation (Section 3.3.1) and the experiment registry (Section 4.2) show which combinations of settings each experiment tests. Section 3.4 gives the exact LLM prompt template.

B Compute Budget

The reported classical training time is approximately 30 seconds on a CPU. The encoder and LLM experiments require GPU resources; Table 11 reports model-specific training and explanation timings. All metrics are computed from saved per-answer predictions and need no GPU.

C Code Availability

The source repository contains the shared data-loading, preprocessing, model and evaluation code. Access to the research data, predictions and model resources needs separate permission. Availability of the source code does not mean that these files are also available. The report's benchmark tables describe the designated final-report version.

D Per-Cell Leaderboards

Experiment outputs are organised by dataset and configuration. Repeating a result requires the matching settings, prediction files and model history. Comparisons using different dataset variants must keep their stated sample sizes. The headline table and frontier breakdown are preserved from the latest final report.

E Frontier-Model Breakdown

Table 13 in Chapter 5 reports five metrics per setting for each frontier model. Table 16 repeats the QWK and exact-mark with the per-model *unparsed* count: responses with no extractable integer score are left out of the metrics rather than given a replacement score. The sample size n therefore differs by row. Gemini has the most unparsed responses

Table 15: Key hyperparameters of each model.

Pillar	Setting	Value
Classical	TF-IDF analyzer / n-grams	`char_wb`, $(2, 4)$
	max features / SVR C / kernel	15,000 / 1.0 / RBF
	extra baselines	`tfidf_cos`, `fasttext_cos` (100-dim)
RNN	vocab / embed / hidden / layers	5,000 / 128 / 128 / 2 (bidirectional)
	dropout / batch / max epochs	0.3 / 64 / 20 (patience 4)
	learning rate / optimiser	$1e{-3}$ / Adam
Transformer	backbones	mBERT, XLM-R, GTE-multilingual
	config / frozen layers / max length / batch	dual + cross / 6 / 256 / 16
	learning rate / weight decay / dropout	$2e{-5}$ / 0.01 / 0.2
LLM	bases / adaptation	Qwen (4B), Gemma (4B), SEA-LION (2.3B) / QLoRA (4-bit NF4, double-quant)
	LoRA rank / alpha / dropout	16 / 16 / 0
	target modules	q, k, v, o, gate, up, down
	epochs / batch (effective) / warm-up	10 / 4 (16 via accumulation) / 5 %
	max sequence length / loss	1,024 / completion-only
	learning rate (val-selected)	$\in \{1, 2, 3\} \times 10^{-4}$
Common	preprocessing modes / input formats	{raw, clean, segment} / {ra, qar}
	max-score feature normaliser	divide by 20 (largest observed max)
	split / seed	70/15/15 / 42 (multi-seed: 42, 1337, 2024, 7, 99)
	target / loss	$\hat{y} \in [0, 1]$ / mean-squared error (regression heads)

in the bare setting, with 65 of 137. We record each exact model version and evaluation date, and save the raw API responses so that the reported snapshot can be checked again.

F Additional Explanation Heatmaps

Figures 15, 16, 17, and 18 show SHAP word-attribution heatmaps for all four pillars on real Khmer answers. We use browser text shaping to display readable Khmer words. Standard plotting tools do not correctly stack subscript consonants or position the vowels.

Table 16: Frontier models, full zero-shot breakdown (bare vs. reasoning) on the `no10c` test set, evaluated 2026-06-24; `exp14` writes `results_frontier/no10c/frontier_metrics.csv`. "Unparsed" counts responses whose integer score could not be extracted; these are excluded from the metrics, so the effective n is 137 minus this count.

Model	Setting	QWK	Exact	Unparsed
GPT-5.5	bare	0.684	0.48	11
GPT-5.5	reasoning	0.708	0.45	1
Claude Opus 4.8	bare	0.662	0.35	0
Claude Opus 4.8	reasoning	0.734	0.43	0
Gemini 3.5 Flash	bare	0.555	0.46	65
Gemini 3.5 Flash	reasoning	0.585	0.45	20
DeepSeek V4	bare	0.553	0.34	3
DeepSeek V4	reasoning	0.579	0.31	1



Figure 16: Sample of a Khmer word-attribution heatmap (BiLSTM model, SHAP).



Figure 17: Sample of a Khmer word-attribution heatmap (Transformer model, SHAP).



Figure 18: Sample of a Khmer word-attribution heatmap (LLM model, SHAP).

G Live Prototype Screenshot

Figure 19 reproduces the interface example shown in the final presentation. It shows the teacher's inputs, model choice, proposed mark, Khmer word highlights and feedback. This is an interface example, not a classroom field study.

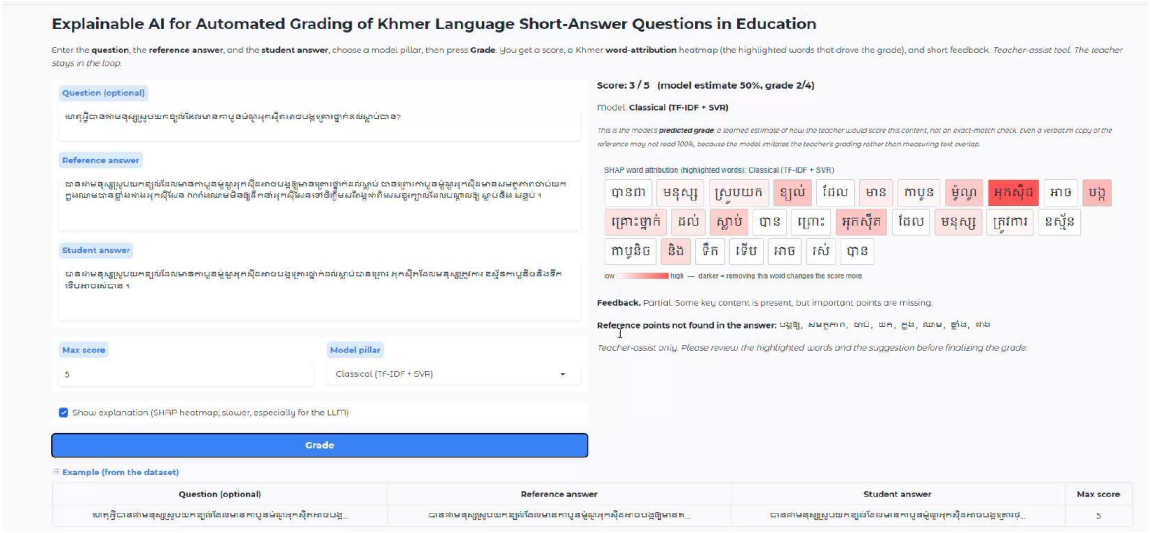


Figure 19: Prototype interface reproduced from the final presentation. The teacher reviews the proposed mark and explanation before deciding the final grade.

H Ethics and Data Governance

The corpus contains real, anonymised student answers from minors, collected from two Cambodian high schools and used solely for research into a teacher-assist grading tool. Access to the examination scripts was granted by approaching each school and obtaining permission *directly from the school Director*; no formal institutional review board (IRB) process applies at this level in this setting. Only coded identifiers are stored; the data contains no student names. The system is intended to *support*, not replace, a teacher's

judgement (human-in-the-loop). Because a single teacher graded all answers, the models reflect that grader's decisions; results measure agreement with this grader. Removing the group with clear grading inconsistencies reduces part of this problem. Section 3.2.3 openly documents the inconsistent marks in class 10C Biology.